

Basic Structures: Sets, Functions, Sequences, Sums, and Matrices

Chapter 2

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Chapter Summary

Sets

- The Language of Sets
- Set Operations
- Set Identities

Functions

- Types of Functions
- Operations on Functions
- Computability

Sequences and Summations

- Types of Sequences
- Summation Formulae

Set Cardinality

- Countable Sets

Matrices

- Matrix Arithmetic

Sets

Section 2.1

Section Summary

Definition of sets

Describing Sets

- Roster Method
- Set-Builder Notation

Some Important Sets in Mathematics

Empty Set and Universal Set

Subsets and Set Equality

Cardinality of Sets

Tuples

Cartesian Product

2.1 Sets

A *set* is an *unordered* collection of objects.

- the students in this class
- the chairs in this room

The objects in a set are called the *elements*, or *members* of the set. A set is said to *contain* its elements.

The notation $a \in A$ denotes that a is an element of the set A .

If a is not a member of A , write $a \notin A$

Describing a Set: Roster Method

$$S = \{a, b, c, d\}$$

Order not important

$$S = \{a, b, c, d\} = \{b, c, a, d\}$$

Each distinct object is either a member or not; listing **more than once** does not change the set.

$$S = \{a, b, c, d\} = \{a, b, c, b, c, d\}$$

Ellipses (...) may be used to describe a set **without listing all of the members** when the pattern is clear.

$$S = \{a, b, c, d, \dots, z\}$$

Roster Method

Set of all vowels in the English alphabet:

$$V = \{a, e, i, o, u\}$$

Set of all odd positive integers less than 10:

$$O = \{1, 3, 5, 7, 9\}$$

Set of all positive integers less than 100:

$$S = \{1, 2, 3, \dots, 99\}$$

Set of all integers less than 0:

$$S = \{\dots, -3, -2, -1\}$$

Some Important Sets

N = *natural numbers* = $\{0,1,2,3,\dots\}$

Z = *integers* = $\{\dots,-3,-2,-1,0,1,2,3,\dots\}$

Z⁺ = *positive integers* = $\{1,2,3,\dots\}$

R = *set of real numbers*

R⁺ = *set of positive real numbers*

C = *set of complex numbers.*

Q = *set of rational numbers*

Set-Builder Notation

Specify the property or properties that all members must satisfy:

$$S = \{x \mid x \text{ is a positive integer less than } 100\}$$

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

$$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$$

A predicate may be used: $S = \{x \mid P(x)\}$

Example: $S = \{x \mid \text{Prime}(x)\}$

Positive rational numbers:

$$\mathbf{Q}^+ = \{x \in \mathbf{R} \mid x = p/q, \text{ for some positive integers } p, q\}$$

Interval Notation

$a, b \in \mathbf{R}$, and $a < b$ then

- $(a, b) = \{x \in \mathbf{R} \mid a < x < b\}$
- $[a, b] = \{x \in \mathbf{R} \mid a \leq x \leq b\}$
- $(a, b] = \{x \in \mathbf{R} \mid a < x \leq b\}$
- $(-\infty, b] = \{x \in \mathbf{R} \mid x \leq b\}$
- $[a, \infty) = \{x \in \mathbf{R} \mid a \leq x\}$
- $(a, \infty) = \{x \in \mathbf{R} \mid a < x\}$

closed interval $[a,b]$

open interval (a,b)

Universal Set and Empty Set

The *universal set* U is the set containing everything currently under consideration.

- Sometimes implicit
- Sometimes explicitly stated.
- Contents depend on the context.

The *empty set (null set)* is the set with no elements.

- $\emptyset$
- $\{\}$

Some things to remember

Sets can be elements of sets.

$$\{\{1,2,3\}, a, \{b,c\}\}$$

$$\{\mathbf{N}, \mathbf{Z}, \mathbf{Q}, \mathbf{R}\}$$

A set with one element is called a singleton set.

The empty set is different from a set containing the empty set.

$$\emptyset \neq \{ \emptyset \}$$

Set Equality

Definition: Two sets are *equal* if and only if they have the same elements.

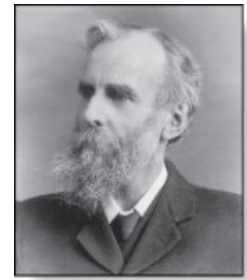
- Therefore if A and B are sets, then A and B are equal if and only if $\forall x(x \in A \leftrightarrow x \in B)$
- We write $A = B$ if A and B are equal sets.

$$\{1, 3, 5\} = \{3, 5, 1\}$$

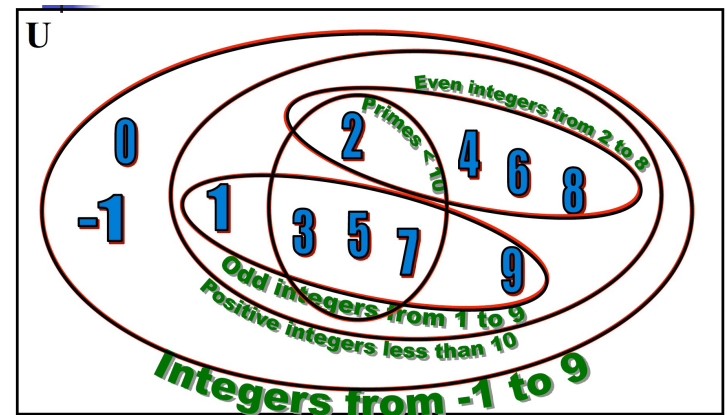
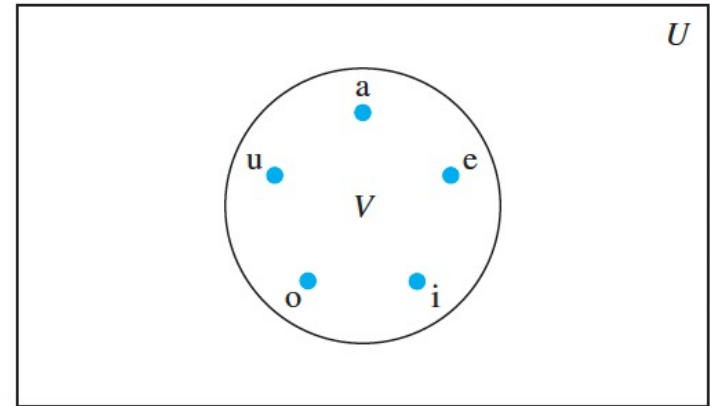
$$\{1, 5, 5, 5, 3, 3, 1\} = \{1, 3, 5\}$$

2.1.2 Venn Diagrams

John Venn (1834-1923)
Cambridge, UK



- In Venn diagrams the universal set U , which contains all the objects under consideration, is represented by a **rectangle**.
- Inside this rectangle, **circles** or other geometrical figures are used to represent sets.
- Sometimes **points** are used to represent the particular elements of the set.
- Venn diagrams are often used to indicate the relationships between sets.



2.1.3 Subsets

Definition: The set A is a *subset* of B , if and only if every element of A is also an element of B .

- The notation $A \subseteq B$ (or $B \supseteq A$) is used to indicate that A is a subset of the set B .
- $A \subseteq B$ holds if and only if $\forall x(x \in A \rightarrow x \in B)$ is true.
- Because $a \in \emptyset$ is always false, $\emptyset \subseteq S$, for every set S .
- Because $a \in S \rightarrow a \in S$, $S \subseteq S$, for every set S .

Showing a Set is or is not a Subset of Another Set

Showing that A is a Subset of B :

To show that $A \subseteq B$, show that if x belongs to A , then x also belongs to B .

Showing that A is not a Subset of B :

To show that A is not a subset of B , $A \not\subseteq B$, find an element $x \in A$ with $x \notin B$. (Such an x is a counterexample to the claim that $x \in A$ implies $x \in B$.)

Examples:

1. The set of all computer science majors at your school is a subset of all students at your school.
2. The set of integers with squares less than 100 is not a subset of the set of nonnegative integers.

Another look at Equality of Sets

Recall that two sets A and B are *equal*, denoted by $A = B$, iff

$$\forall x (x \in A \leftrightarrow x \in B)$$

Using logical equivalences we have that $A = B$ iff

$$\forall x [(x \in A \rightarrow x \in B) \wedge (x \in B \rightarrow x \in A)]$$

This is equivalent to

$$A \subseteq B \quad \text{and} \quad B \subseteq A$$

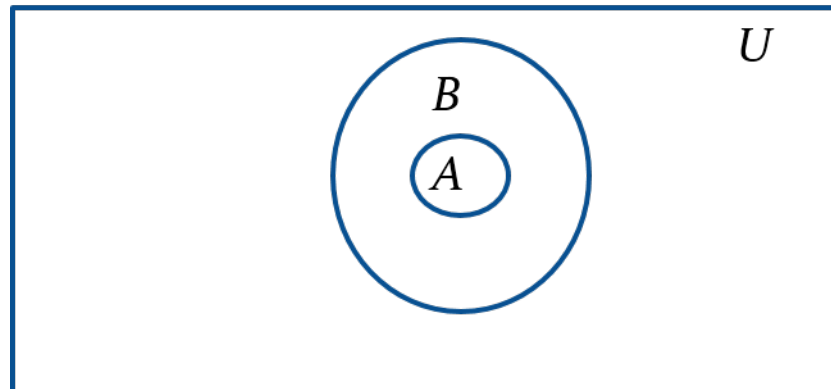
Proper Subsets

Definition: If $A \subseteq B$, but $A \neq B$, then we say A is a *proper subset* of B , denoted by $A \subset B$. If $A \subset B$, then

$$\forall x(x \in A \rightarrow x \in B) \wedge \exists x(x \in B \wedge x \notin A)$$

is true.

Venn Diagram



2.1.4 The Size of a Set

Definition: If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is *finite*. Otherwise it is *infinite*.

Definition: The *cardinality* of a finite set A , denoted by $|A|$, is the number of (distinct) elements of A .

Examples:

1. $|\emptyset| = 0$
2. Let S be the letters of the English alphabet. Then $|S| = 26$
3. $|\{1,2,3\}| = 3$
4. $|\{\emptyset\}| = 1$
5. The set of integers is infinite.

2.1.5 Power Sets

Definition: The set of all subsets of a set A , denoted $\mathcal{P}(A)$, is called the *power set* of A .

Example: If $A = \{a, b\}$ then

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$

If a set has n elements, then the cardinality of the power set is 2^n . (In Chapters 5 and 6, we will discuss different ways to show this.)

2.1.5 Power Sets

Examples

- $\mathcal{P}(\{a, b\}) =$
 $\{ \emptyset, \{a\}, \{b\}, \{a, b\} \}$
- $\mathcal{P}(\{0, 1, 2\}) =$
 $\{\emptyset, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$
- $\mathcal{P}(\emptyset) =$
 $\{\emptyset\}$
- $\mathcal{P}(\{\emptyset\}) =$
 $\{\emptyset, \{\emptyset\}\}$

2.1.6 Cartesian Product: Tuples

The *ordered n -tuple* $(a_1, a_2, \dots, a_n)$ is the *ordered* collection that has a_1 as its first element and a_2 as its second element and so on until a_n as its last element.

We say that *two ordered n -tuples are equal* if and only if each corresponding pair of their elements is equal.

In other words, $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ if and only if $a_i = b_i$, for $i = 1, 2, \dots, n$.

2.1.6 Cartesian Product: Tuples

In particular, **ordered 2-tuples** are equal if and only if their corresponding elements are equal.

2-tuples are called *ordered pairs*.

The ordered pairs (a,b) and (c,d) are equal if and only if $a = c$ and $b = d$.

2.1.6 Cartesian Product₁

René Descartes
(1596-1650)

Definition: The *Cartesian Product* of two sets A and B , denoted by $A \times B$ is the set of **ordered pairs** (a,b) where $a \in A$ and $b \in B$.

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$



Example:

$$A = \{a, b\} \quad B = \{1, 2, 3\}$$

$$A \times B = \{(a, 1), (a, 2), (a, 3), (b, 1), (b, 2), (b, 3)\}$$

Definition: The subset R of the Cartesian product $A \times B$ is called a *relation* from the set A to the set B . (Relations will be covered in depth in Chapter 9.)

Cartesian Product₂

Definition: The Cartesian products of the sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$ where a_i belongs to A_i for $i = 1, \dots, n$.

$$A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$$

Example: What is $A \times B \times C$ where $A = \{0,1\}$, $B = \{1,2\}$ and $C = \{0,1,2\}$

Solution:

$$A \times B \times C = \{(0,1,0), (0,1,1), (0,1,2), (0,2,0), (0,2,1), (0,2,2), (1,1,0), (1,1,1), (1,1,2), (1,2,0), (1,2,1), (1,2,2)\}$$

2.1.8 Truth Sets of Quantifiers

Given a **predicate** P and a **domain** D , we define the **truth set of P** to be the set of elements in D for which $P(x)$ is true. The **truth set of $P(x)$** is denoted by

$$\{x \in D \mid P(x)\}$$

What are the truth sets of the predicates $P(x)$, $Q(x)$, and $R(x)$, where the domain is the set of integers and $P(x)$ is “ $|x| = 1$,” $Q(x)$ is “ $x^2 = 2$,” and $R(x)$ is “ $|x| = x$.”

P is the set $\{-1, 1\}$.

Q is the $\emptyset$.

R is $\mathbb{N}$ (the set of nonnegative integers).

2.1.8 Truth Sets of Quantifiers

$\forall xP(x)$ is true over the domain U if and only if the truth set of P is the set U .

$\exists xP(x)$ is true over the domain U if and only if the truth set of P is nonempty.

Homework after class

2.1 Sets

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Set Operations

Section 2.2

Section Summary

Set Operations

- Union
- Intersection
- Difference
- Complementation

More on Set Cardinality

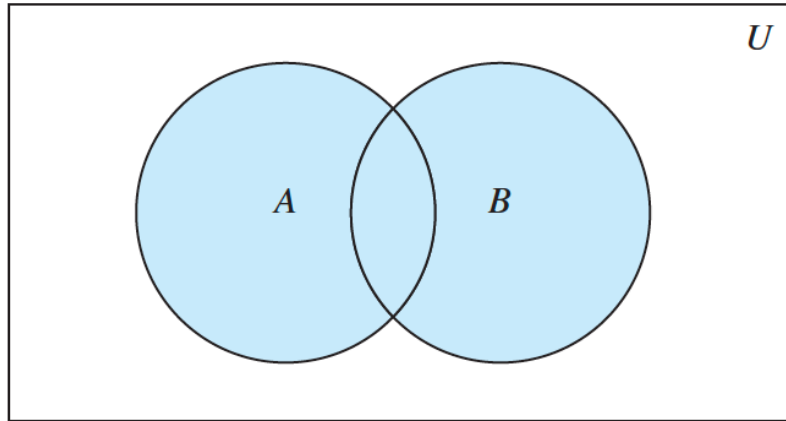
Set Identities

Proving Identities

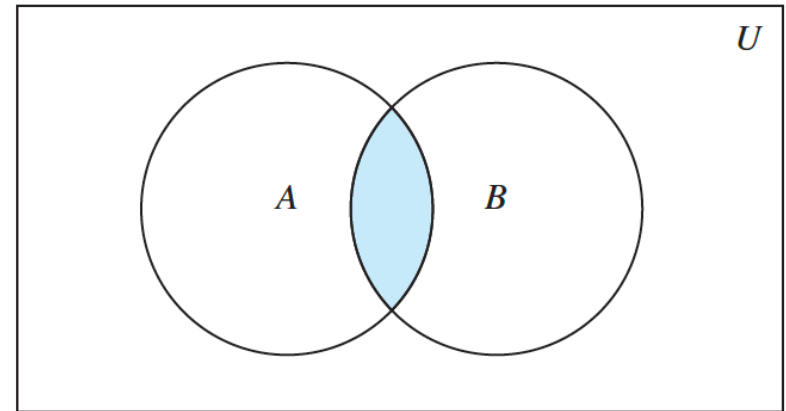
Membership Tables

2.2.1 Introduction

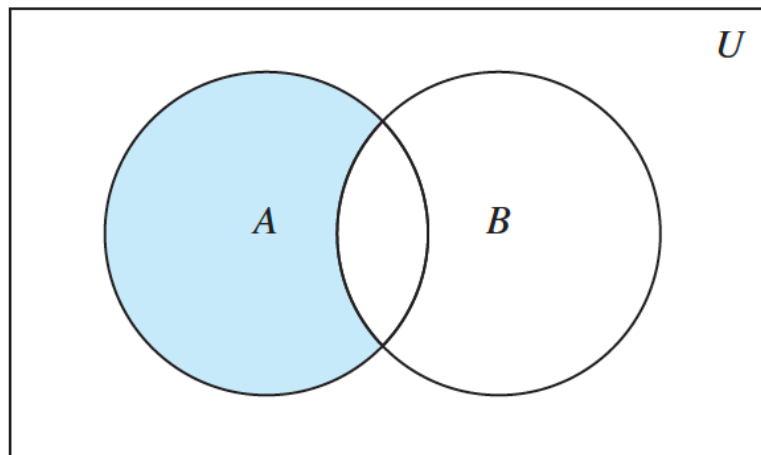
Set Operations



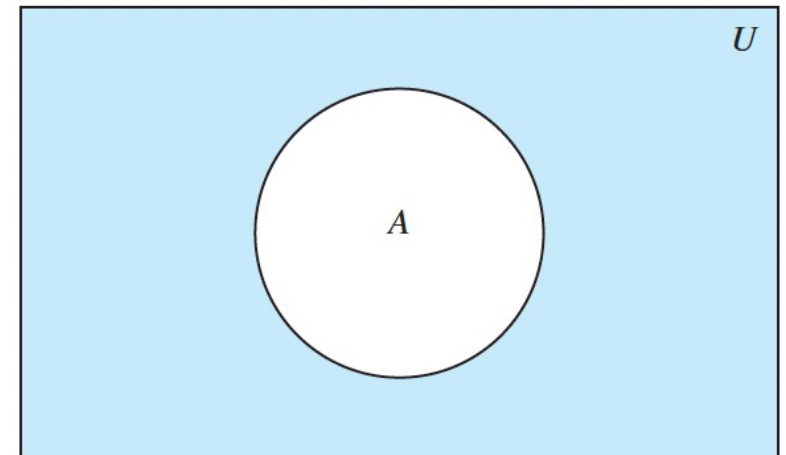
Union $A \cup B$ is shaded.



Intersection $A \cap B$ is shaded.



Difference $A - B$ is shaded.

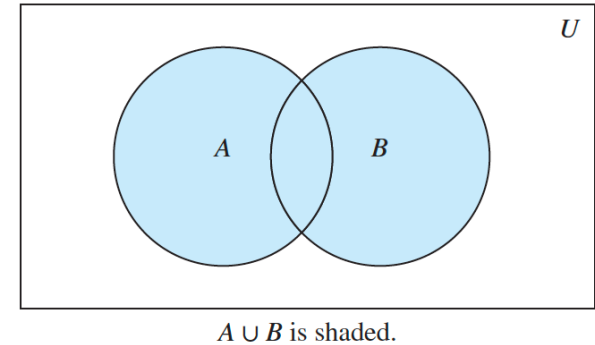


Complementation $\bar{A}$ is shaded.

Union

Definition: Let A and B be sets. The *union* of the sets A and B , denoted by $A \cup B$, is the set:

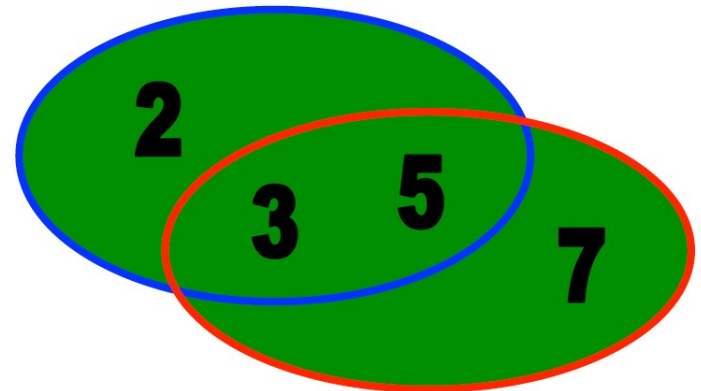
$$\{x \mid x \in A \vee x \in B\}$$



Example: What is $\{2,3,5\} \cup \{3, 5,7\}$?

Solution:

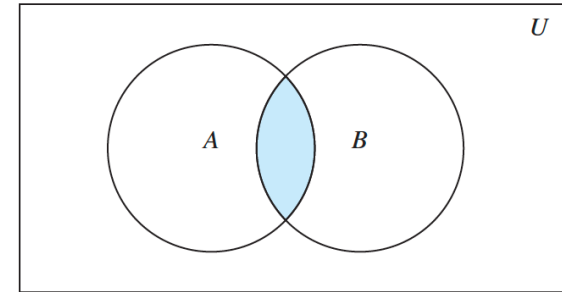
$$\{2,3,5,7\}$$



Intersection

Definition: The *intersection* of sets A and B , denoted by $A \cap B$, is

$$\{x \mid x \in A \wedge x \in B\}$$



$A \cap B$ is shaded.

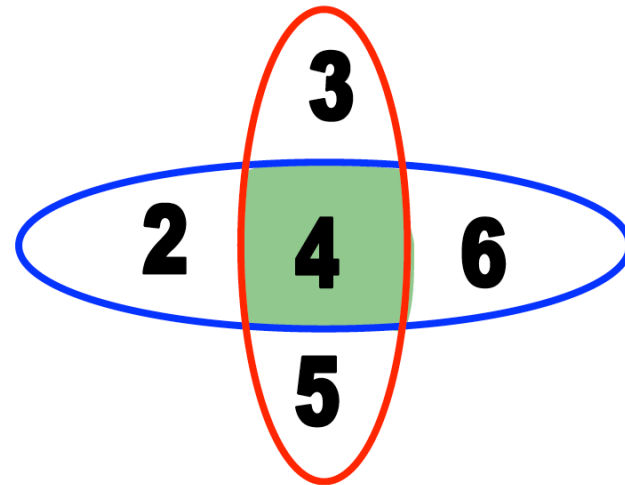
Note if the intersection is empty, then A and B are said to be *disjoint*.

Example: $\{2,4,6\} \cap \{3,4,5\}$?

Solution: $\{4\}$

Example: $\{1,2,3\} \cap \{4,5,6\}$?

Solution: $\emptyset$



Difference

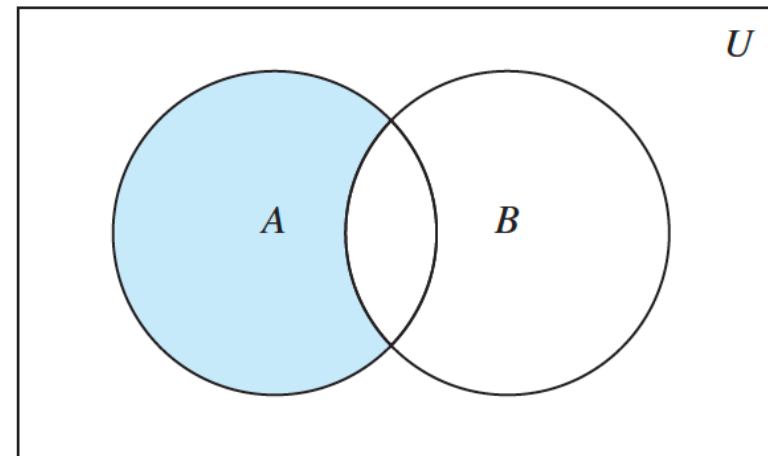
Definition: Let A and B be sets. The *difference* of A and B , denoted by $A - B$ (or $A \setminus B$), is the set containing the elements of A that are not in B . The difference of A and B is also called the complement of B with respect to A .

$$A - B = \{x | x \in A \wedge x \notin B\} = A \cap \overline{B}$$

Example: $\{1,2,3\} - \{3, 4, 5\}$?

Solution:

$\{1,2\}$



$A - B$ is shaded.

Complement

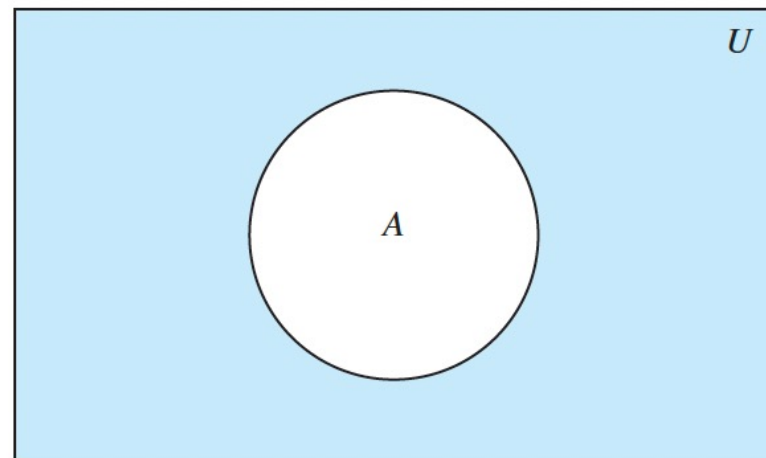
Definition: If A is a set, then the *complement* of the A (with respect to U), denoted by $\bar{A}$ is the set $U - A$

$$\bar{A} = \{x | x \in U | x \notin A\}$$

$$\bar{A} = \{x \in U \mid x \notin A\}$$

Example: If U is the positive integers less than 100, what is the complement of $\{x | x > 70\}$

Solution: $\{x | x \leq 70\}$

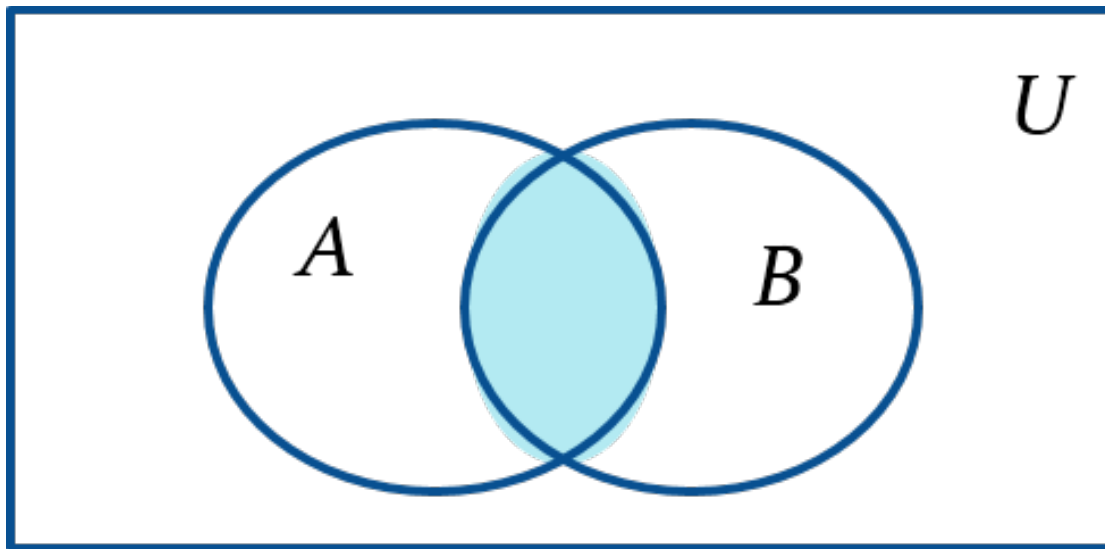


$\bar{A}$ is shaded.

The Cardinality of the Union of Two Sets

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Venn Diagram for A , B , $A \cap B$, $A \cup B$



Review Questions

$$U = \{0,1,2,3,4,5,6,7,8,9,10\}$$

$$A = \{1,2,3,4,5\}, \quad B = \{4,5,6,7,8\}$$

- $A \cup B$ $\{1,2,3,4,5,6,7,8\}$
- $A \cap B$ $\{4,5\}$
- $\bar{A}$ $\{0,6,7,8,9,10\}$
- $\bar{B}$ $\{0,1,2,3,9,10\}$
- $A - B$ $\{1,2,3\}$
- $B - A$ $\{6,7,8\}$

Symmetric Difference (*optional*)

Definition: The *symmetric difference* of **A** and **B**, denoted by $A \oplus B$ is the set

$$(A - B) \cup (B - A)$$

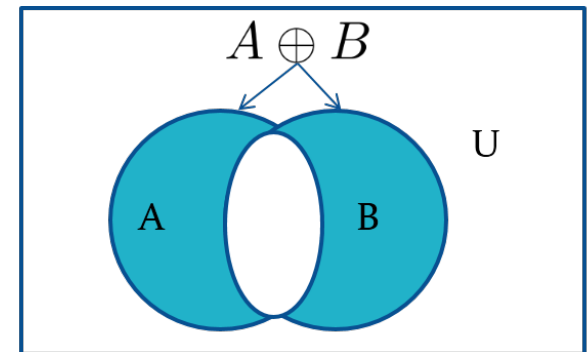
Example:

$$U = \{0,1,2,3,4,5,6,7,8,9,10\}$$

$$A = \{1,2,3,4,5\} \quad B = \{4,5,6,7,8\}$$

What is $A \oplus B$:

Solution: $\{1,2,3,6,7,8\}$



Venn Diagram

2.2.2 Set Identities

- Identity: $A \cup \emptyset = A = A \cap U$
- Domination: $A \cup U = U, A \cap \emptyset = \emptyset$
- Idempotent: $A \cup A = A, A \cap A = A$
- Double complement: $\overline{\overline{A}} = A$
- Commutative: $A \cup B = B \cup A, A \cap B = B \cap A$
- Associative: $A \cup (B \cup C) = (A \cup B) \cup C,$
 $A \cap (B \cap C) = (A \cap B) \cap C$
- Distributive: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C),$
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- Absorption: $A \cup (A \cap B) = A, A \cap (A \cup B) = A$
- Complement: $A \cup \overline{A} = U, A \cap \overline{A} = \emptyset$
- De Morgan's $\overline{A \cup B} = \overline{A} \cap \overline{B} \quad \overline{A \cap B} = \overline{A} \cup \overline{B}$

Proving Set Identities

Different ways to **prove** set identities:

- **Subset method**: Show that each side of the identity is a subset of the other side.
- **Membership table**: For each possible combination of the atomic sets, show that an element in exactly these atomic sets must either belong to both sides or belong to neither side.
- **Apply existing identities**: Start with one side, transform it into the other side using a sequence of steps by applying an established identity.
- **Venn Diagram**

Way 1 : Subset method₁

Example: Prove that De Morgan Law $\overline{A \cap B} = \overline{A} \cup \overline{B}$

Solution: We prove this identity by showing that:

$$1) \overline{A \cap B} \subseteq \overline{A} \cup \overline{B} \text{ and}$$

$$2) \overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$$

Way 1 : Subset method₂

These steps (1) show that: $\overline{A \cap B} \subseteq \overline{A} \cup \overline{B}$

$x \in \overline{A \cap B}$ by assumption

$x \notin A \cap B$ defn. of complement

$\neg((x \in A) \wedge (x \in B))$ by defn. of intersection

$\neg(x \in A) \vee \neg(x \in B)$ 1st De Morgan law for Prop Logic

$x \notin A \vee x \notin B$ defn. of negation

$x \in \overline{A} \vee x \in \overline{B}$ defn. of complement

$x \in \overline{A} \cup \overline{B}$ by defn. of union

Way 1 : Subset method₃

These steps (2) show that: $\overline{A} \cup \overline{B} \subseteq \overline{A \cap B}$

$$x \in \overline{A} \cup \overline{B}$$

by assumption

$$(x \in \overline{A}) \vee (x \in \overline{B})$$

by defn. of union

$$(x \notin A) \vee (x \in \overline{B})$$

defn. of complement

$$\neg(x \in A) \vee \neg(x \in B)$$

defn. of negation

$$\neg((x \in A) \wedge (x \in B))$$

1st De Morgan law for Prop Logic

$$\neg(x \in A \cap B)$$

defn. of intersection

$$x \in \overline{A \cap B}$$

defn. of complement

Way 2 : Set-Builder Notation

$$O = \{x \in \mathbf{Z}^+ \mid x \text{ is odd and } x < 10\}$$

$$\overline{A \cap B} = \bar{A} \cup \bar{B}.$$

$\overline{A \cap B} = \{x \mid x \notin A \cap B\}$	by definition of complement
$= \{x \mid \neg(x \in (A \cap B))\}$	by definition of does not belong symbol
$= \{x \mid \neg(x \in A \wedge x \in B)\}$	by definition of intersection
$= \{x \mid \neg(x \in A) \vee \neg(x \in B)\}$	by the first De Morgan law for logical equivalences
$= \{x \mid x \notin A \vee x \notin B\}$	by definition of does not belong symbol
$= \{x \mid x \in \bar{A} \vee x \in \bar{B}\}$	by definition of complement
$= \{x \mid x \in \bar{A} \cup \bar{B}\}$	by definition of union
$= \bar{A} \cup \bar{B}$	by meaning of set builder notation

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

<i>A</i>	<i>B</i>	<i>C</i>	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1					
1	1	0					
1	0	1					
1	0	0					
0	1	1					
0	1	0					
0	0	1					
0	0	0					

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

<i>A</i>	<i>B</i>	<i>C</i>	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1				
1	1	0	0				
1	0	1	0				
1	0	0	0				
0	1	1	1				
0	1	0	0				
0	0	1	0				
0	0	0	0				

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

<i>A</i>	<i>B</i>	<i>C</i>	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1			
1	1	0	0	1			
1	0	1	0	1			
1	0	0	0	1			
0	1	1	1	1			
0	1	0	0	0			
0	0	1	0	0			
0	0	0	0	0			

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

<i>A</i>	<i>B</i>	<i>C</i>	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1		
1	1	0	0	1	1		
1	0	1	0	1	1		
1	0	0	0	1	1		
0	1	1	1	1	1		
0	1	0	0	0	1		
0	0	1	0	0	0		
0	0	0	0	0	0		

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

<i>A</i>	<i>B</i>	<i>C</i>	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	
1	1	0	0	1	1	1	
1	0	1	0	1	1	1	
1	0	0	0	1	1	1	
0	1	1	1	1	1	1	
0	1	0	0	0	1	0	
0	0	1	0	0	0	1	
0	0	0	0	0	0	0	

Way 3 : Membership Table

Example: Construct a membership table to show that the distributive law holds.

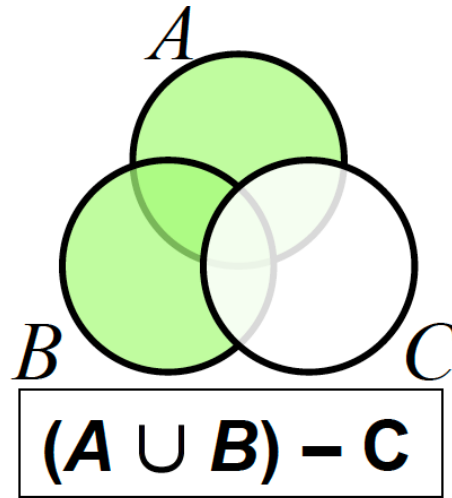
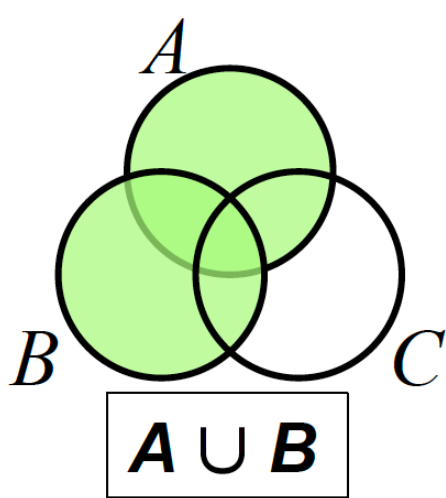
$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Solution:

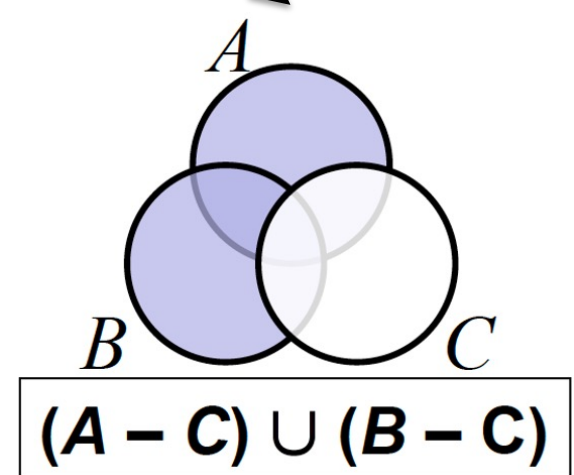
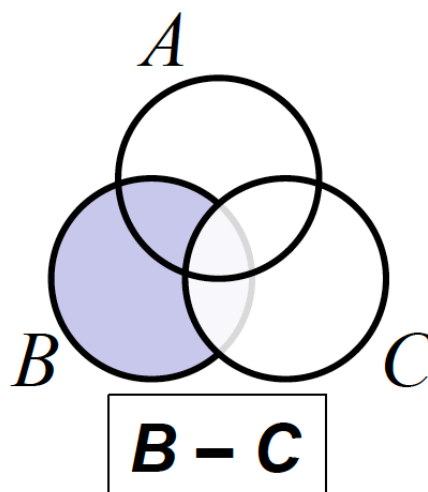
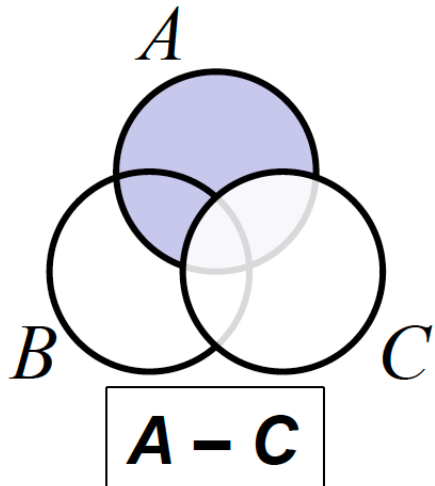
<i>A</i>	<i>B</i>	<i>C</i>	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

Way 4 : Venn Diagram

Prove $(A \cup B) - C = (A - C) \cup (B - C)$.



Same



Methods of Proving Set Identities

TABLE 3 Methods of Proving Set Identities.

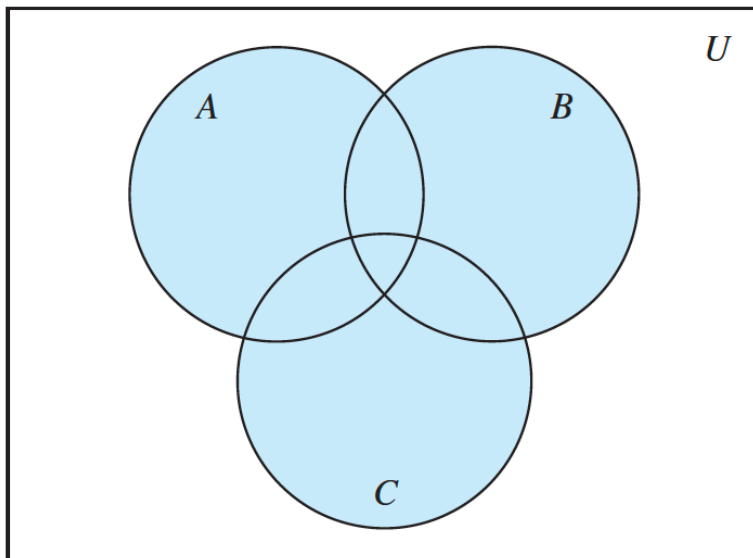
<i>Description</i>	<i>Method</i>
Subset method	Show that each side of the identity is a subset of the other side.
Membership table	For each possible combination of the atomic sets, show that an element in exactly these atomic sets must either belong to both sides or belong to neither side
Apply existing identities	Start with one side, transform it into the other side using a sequence of steps by applying an established identity.

Venn Diagram

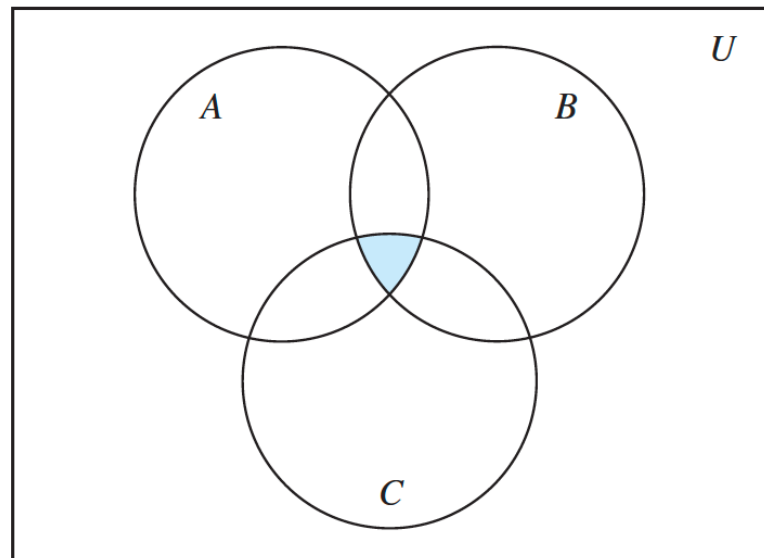
2.2.3 Generalized Unions and Intersections

Because unions and intersections of sets satisfy associative laws, the sets $A \cup B \cup C$ and $A \cap B \cap C$ are well defined (unambiguous);

That is, we do not have to use parentheses to indicate which operation comes first because $A \cup (B \cup C) = (A \cup B) \cup C$ and $A \cap (B \cap C) = (A \cap B) \cap C$.



(a) $A \cup B \cup C$ is shaded.



(b) $A \cap B \cap C$ is shaded.

2.2.3 Generalized Unions and Intersections

Let $A_1, A_2, \dots, A_n$ be an indexed collection of sets.

We define:

$$A_1 \cup A_2 \cup \dots \cup A_n = \bigcup_{i=1}^n A_i$$

$$A_1 \cap A_2 \cap \dots \cap A_n = \bigcap_{i=1}^n A_i$$

Example: For $i = 1, 2, \dots$, let $A_i = \{i, i + 1, i + 2, \dots\}$. Then,

$$\bigcup_{i=1}^n A_i = \bigcup_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{1, 2, 3, \dots\}$$

$$\bigcap_{i=1}^n A_i = \bigcap_{i=1}^n \{i, i + 1, i + 2, \dots\} = \{n, n + 1, n + 2, \dots\} = A_n$$

2.2.4 Computer Representation of Sets

- For an enumerable universal set U with ordering $x_1, x_2, x_3, \dots$, we can represent a finite set $S \subseteq U$ as the finite bit string $B = b_1b_2\dots b_n$ where $b_i = 1$ if $x_i \in S$ and $b_i = 0$ if $x_i \notin S$.
- Example. $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
 - The subset of all even integers in U , namely, $\{2, 4, 6, 8, 10\}$, by the string: **01 0101 0101**.
 - The set of all integers in U that do not exceed 5, namely, $\{1, 2, 3, 4, 5\}$, is represented by the string: **11 1110 0000**.
- In this representation, the set operators “ $\cup$ ”, “ $\cap$ ”, “ $-$ ” are implemented directly by bitwise OR, AND, NOT!

2.2.4 Computer Representation of Sets

Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, and the ordering of elements of U has the elements in increasing order, then

$$S1 = \{1, 2, 3, 4, 5\} \Rightarrow B1 = 11\ 1110\ 0000$$

$$S2 = \{1, 3, 5, 7, 9\} \Rightarrow B2 = 10\ 1010\ 1010$$

- $S1 \cup S2 = \{1, 2, 3, 4, 5, 7, 9\} \Rightarrow$ bit string =

$$11\ 1110\ 1010 = B1 \vee B2$$

- $S1 \cap S2 = \{1, 3, 5\} \Rightarrow$ bit string =

$$10\ 1010\ 0000 = B1 \wedge B2$$

- $\bar{S}_1 = \{6, 7, 8, 9, 10\} \Rightarrow$ bit string =

$$00\ 0001\ 1111 = \neg B1$$

2.2.5 Multisets

Sometimes the **number of times** that an element occurs in an unordered collection matters.

A **multiset** (short for multiple-membership set) is an **unordered** collection of elements where an element can occur as **a member more than once**.

We can use the **same notation** for a multiset as we do for a set, but each element is **listed the number of times** it occurs.

The multiset denoted by $\{a, a, a, b, b\}$ is the multiset that contains the element a thrice and the element b twice.

2.2.5 Multisets

We can avoid this ambiguity by using an alternate notation for multisets. The notation $\{m_1 \cdot a_1, m_2 \cdot a_2, \dots, m_r \cdot a_r\}$ denotes the multiset with element a_1 occurring m_1 times, element a_2 occurring m_2 times, and so on.

The numbers m_i , $i = 1, 2, \dots, r$, are called the **multiplicities** of the elements a_i , $i = 1, 2, \dots, r$.

Elements not in a multiset are assigned **0** as their multiplicity in this set.

The **cardinality** of a multiset is defined to be the sum of the multiplicities of its elements.

2.2.5 Multisets

Suppose that P and Q are the multisets $\{4 \cdot a, 1 \cdot b, 3 \cdot c\}$ and $\{3 \cdot a, 4 \cdot b, 2 \cdot d\}$, respectively. Find $P \cup Q$, $P \cap Q$, $P - Q$, and $P + Q$.

$$\begin{aligned} P \cup Q &= \{\max(4, 3) \cdot a, \max(1, 4) \cdot b, \max(3, 0) \cdot c, \max(0, 2) \cdot d\} \\ &= \{4 \cdot a, 4 \cdot b, 3 \cdot c, 2 \cdot d\}, \end{aligned}$$

$$\begin{aligned} P \cap Q &= \{\min(4, 3) \cdot a, \min(1, 4) \cdot b, \min(3, 0) \cdot c, \min(0, 2) \cdot d\} \\ &= \{3 \cdot a, 1 \cdot b, 0 \cdot c, 0 \cdot d\} = \{3 \cdot a, 1 \cdot b\}, \end{aligned}$$

$$\begin{aligned} P - Q &= \{\max(4 - 3, 0) \cdot a, \max(1 - 4, 0) \cdot b, \max(3 - 0, 0) \cdot c, \max(0 - 2, 0) \cdot d\} \\ &= \{1 \cdot a, 0 \cdot b, 3 \cdot c, 0 \cdot d\} = \{1 \cdot a, 3 \cdot c\}, \text{ and} \end{aligned}$$

$$\begin{aligned} P + Q &= \{(4 + 3) \cdot a, (1 + 4) \cdot b, (3 + 0) \cdot c, (0 + 2) \cdot d\} \\ &= \{7 \cdot a, 5 \cdot b, 3 \cdot c, 2 \cdot d\}. \end{aligned}$$

Section Summary

Set Operations

- Union
- Intersection
- Difference
- Complementation

More on Set Cardinality

Set Identities

Proving Identities

Membership Tables

Homework after class

2.2 Set Operations

- 20 (a, and c)
- 27
- 63

Functions

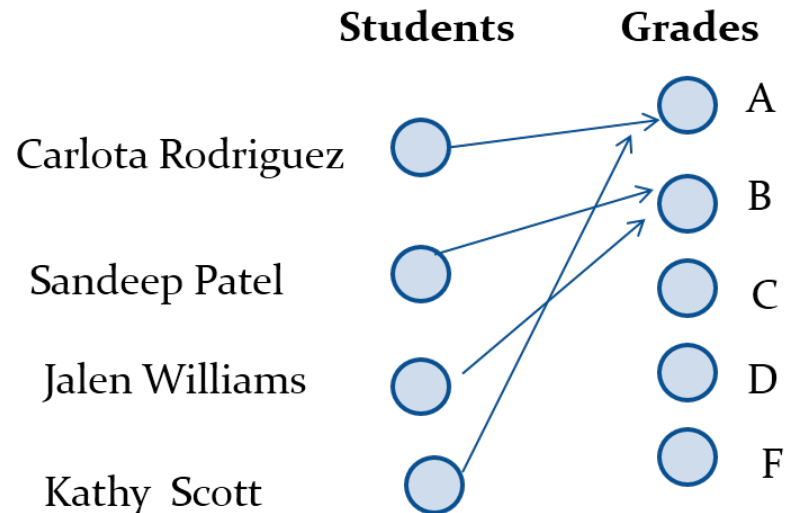
Section 2.3

2.3.1 Functions₁

Definition: Let A and B be nonempty sets. A *function* f from A to B , denoted $f: A \rightarrow B$ is an assignment of each element of A to exactly one element of B .

We write $f(a) = b$ if b is the **unique element of B** assigned by the function f to the element a of A .

- Functions are sometimes called *mapping* or *transformation*.



Representing Functions

Functions may be specified in different ways:

- An explicit statement of the assignment.
 - Students and grades example.

- A formula.

$$f(x) = x + 1$$

- A computer program.
 - A Java program that when given an integer n , produces the n th Fibonacci Number (covered in the next section and also in Chapter 5).

Functions₂

A function $f: A \rightarrow B$ can also be defined as a subset of $A \times B$ (a relation).

This subset is restricted to be a relation where no two elements of the relation have the same first element.

Specifically, a function f from A to B contains one, and only one ordered pair (a, b) for every element $a \in A$.

$$\forall x [x \in A \rightarrow \exists y [y \in B \wedge (x, y) \in f]]$$

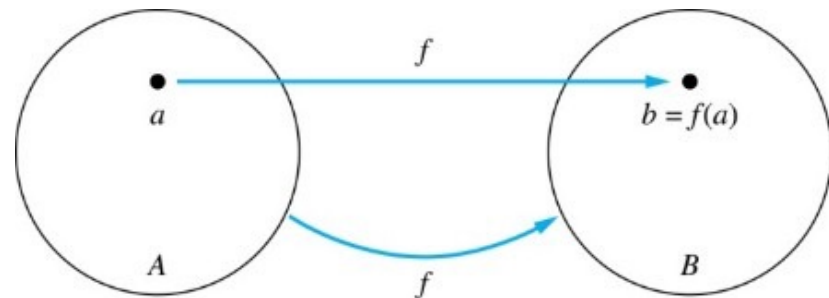
and

$$\forall x, y_1, y_2 [(x, y_1) \in f \wedge (x, y_2) \in f] \rightarrow y_1 = y_2]$$

Functions₃

Given a function $f: A \rightarrow B$:

- We say f *maps* A to B or f is a *mapping* from A to B .
- A is called the *domain* of f .
- B is called the *codomain* of f .
- If $f(a) = b$,
 - then b is called the *image* of a under f .
 - a is called the *preimage* of b .
- The *range* of f is the set of all images of points in A under f . We denote it by $f(A)$.
- Two functions are *equal* *when* they have the same domain, the same codomain and map each element of the domain to the same element of the codomain.



Questions

$f(a) = ?$ z

The image of d is ? z

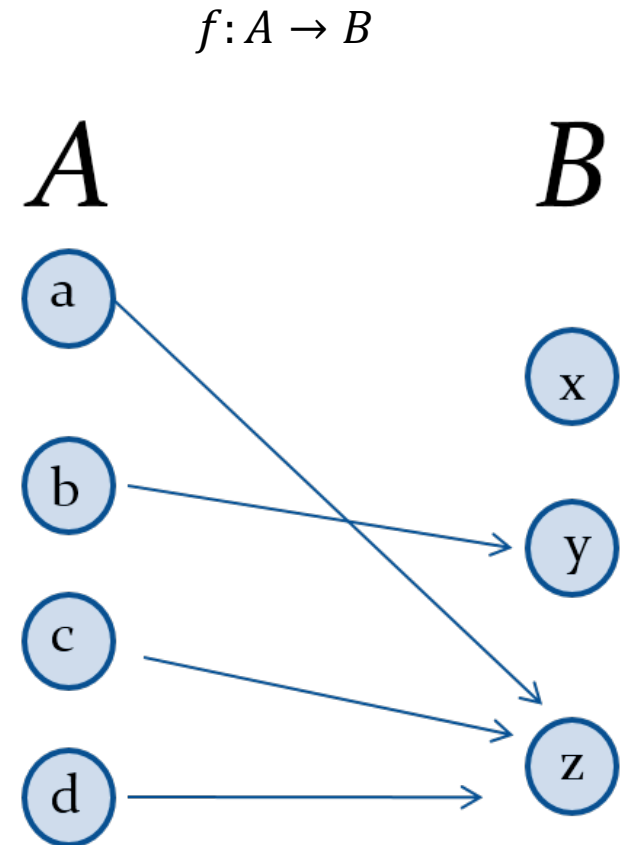
The domain of f is ? A

The codomain of f is ? B

The preimage of y is ? b

$f(A) = ?$ $\{y, z\}$

The preimage(s) of z is (are) ? $\{a, c, d\}$



real-valued and integer-valued

A function is called **real-valued** if its codomain is the set of real numbers.

A function is called **integer-valued** if its codomain is the set of integers.

Two real-valued functions or two integer-valued functions with the same domain can be **added**, as well as **multiplied**.

Let f_1 and f_2 be functions from A to $\mathbf{R}$. Then $f_1 + f_2$ and $f_1 f_2$ are also functions from A to $\mathbf{R}$ defined for all $x \in A$ by

$$(f_1 + f_2)(x) = f_1(x) + f_2(x),$$

$$(f_1 f_2)(x) = f_1(x) f_2(x).$$

Note that the functions $f_1 + f_2$ and $f_1 f_2$ have been defined by specifying their values at x in terms of the values of f_1 and f_2 at x .

real-valued and integer-valued

Let f_1 and f_2 be functions from $\mathbb{R}$ to $\mathbb{R}$ such that $f_1(x) = x^2$ and $f_2(x) = x - x^2$. What are the functions $f_1 + f_2$ and $f_1 f_2$?

Solution: From the definition of the sum and product of functions, it follows that

$$(f_1 + f_2)(x) = f_1(x) + f_2(x) = x^2 + (x - x^2) = x$$

and

$$(f_1 f_2)(x) = x^2(x - x^2) = x^3 - x^4.$$

Image of a subset

When f is a function from A to B , the **image** of a **subset of A** can also be defined.

Let f be a function from A to B and let S be a subset of A . The *image* of S under the function f is the subset of B that consists of the images of the elements of S . We denote the image of S by $f(S)$, so

$$f(S) = \{t \mid \exists s \in S (t = f(s))\}.$$

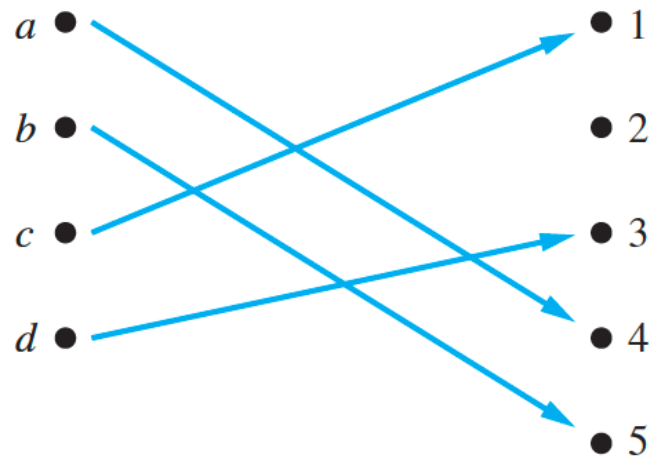
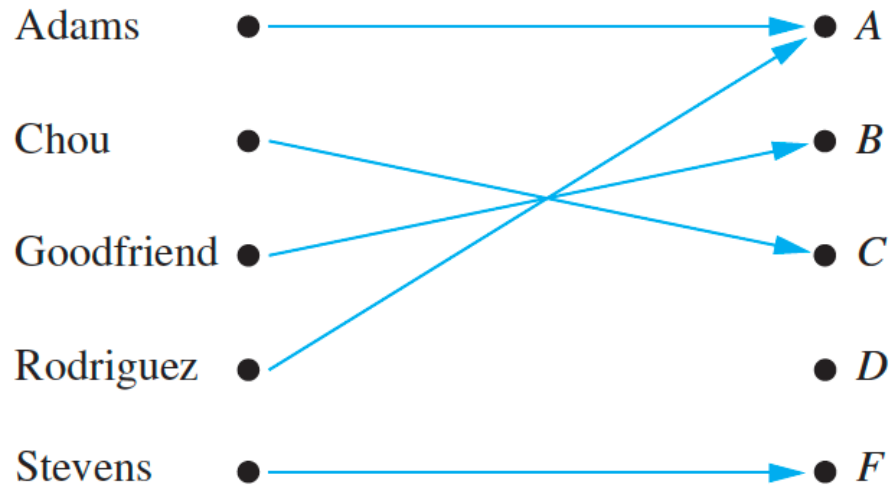
We also use the shorthand $\{f(s) \mid s \in S\}$ to denote this set.

Here, $f(S)$ denotes a set, and not the value of the function f for the set S .

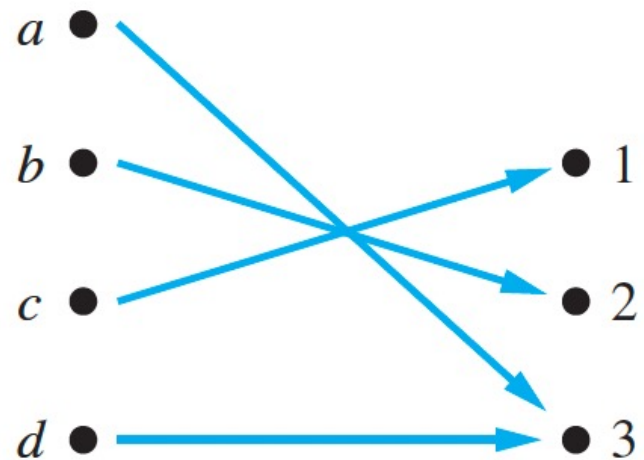
Let $A = \{a, b, c, d, e\}$ and $B = \{1, 2, 3, 4\}$ with $f(a) = 2$, $f(b) = 1$, $f(c) = 4$, $f(d) = 1$, and $f(e) = 1$. What is the image of the subset $S = \{b, c, d\}$?

$$f(S) = \{1, 4\}.$$

2.3.2 One-to-One and Onto Functions



One-to-One

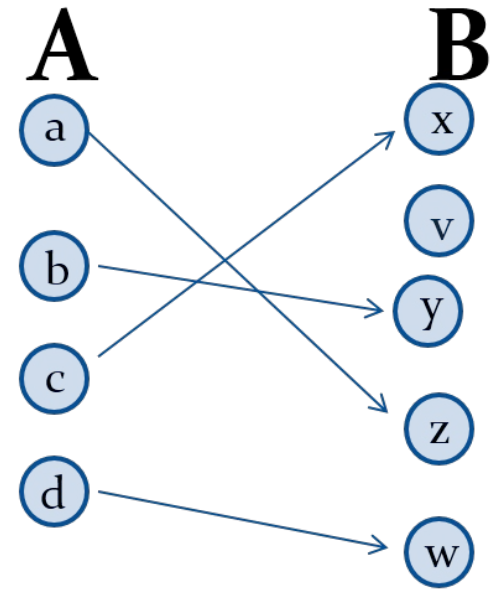


Onto

One-to-One

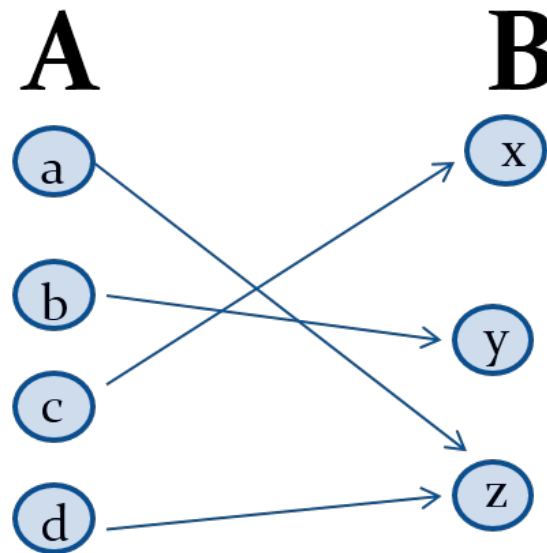
Some functions never assign the same value to two different domain elements. These functions are said to be **one-to-one**.

Definition: A function f is said to be **one-to-one**, or an **injection**, if and only if $f(a) = f(b)$ implies that $a = b$ for all a and b in the domain of f . A function is said to be injective if it is one-to-one.



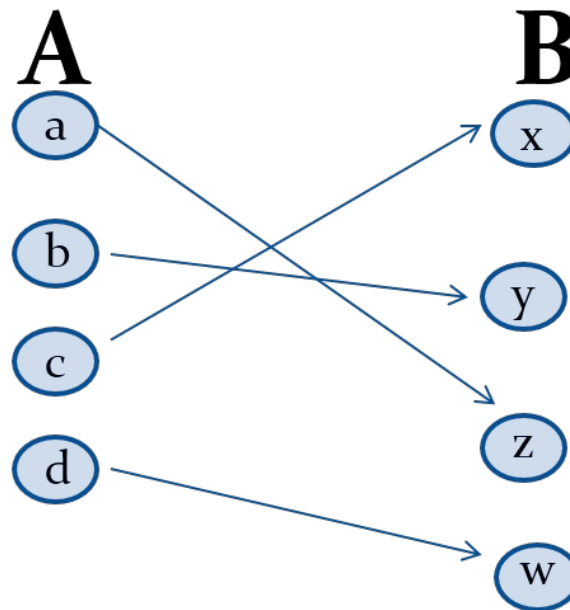
Onto

Definition: A function f from A to B is called *onto* or *surjective*, if and only if for every element $b \in B$ there is an element $a \in A$ with $f(a) = b$.
A function f is called a *surjection* if it is *onto*.

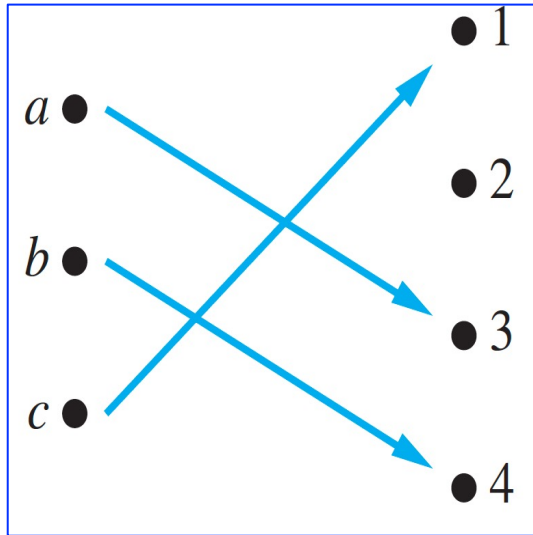


Bijections

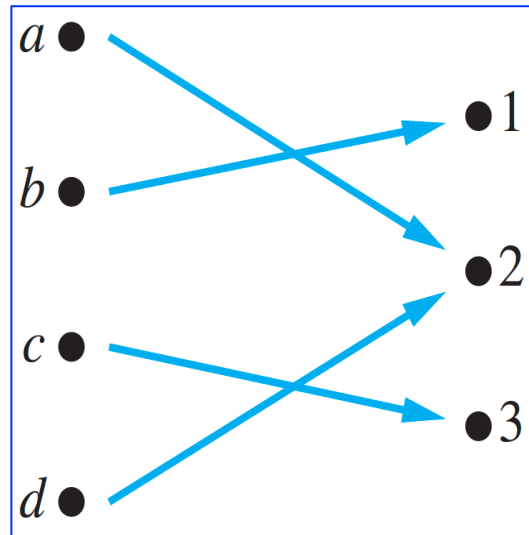
Definition: A function f is a *one-to-one correspondence*, or a *bijection*, if it is both one-to-one and onto (surjective and injective).



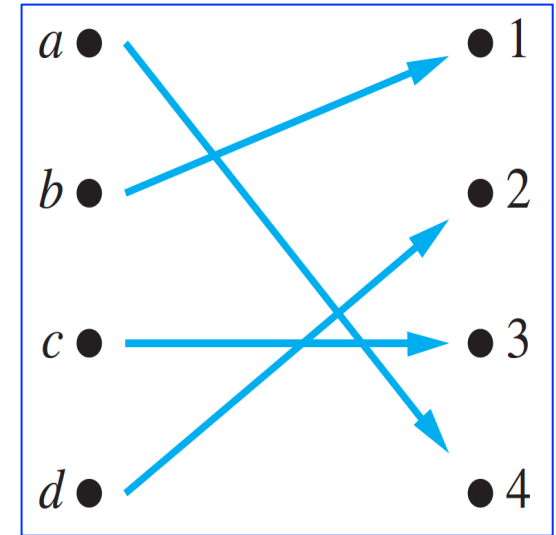
2.3.2 One-to-One and Onto Functions



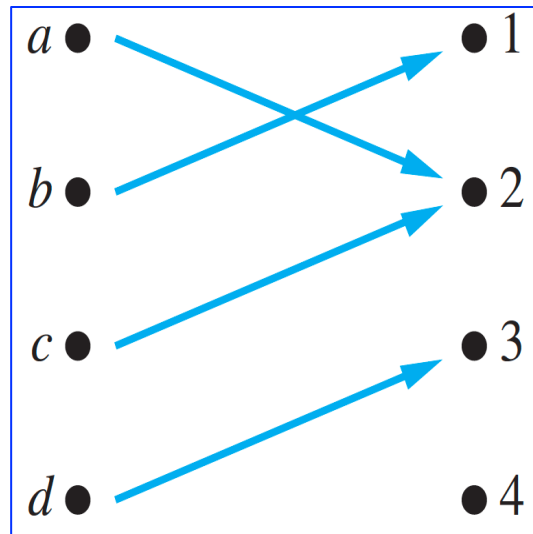
One-to-one, not onto



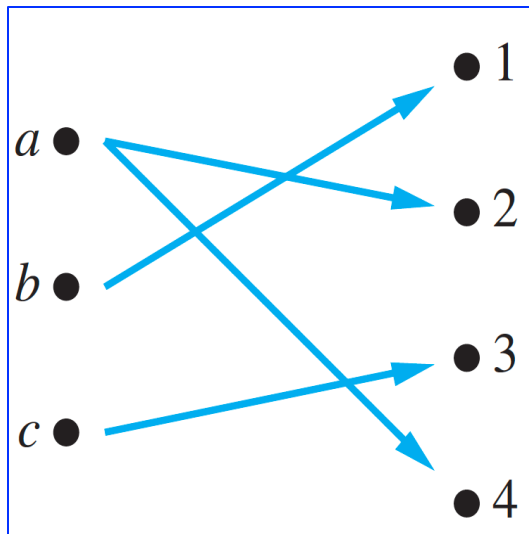
Onto, not one-to-one



One-to-one and onto



Neither one-to-one nor onto



Not a function

Showing that f is one-to-one or onto₁

Suppose that $f : A \rightarrow B$.

- *To show that f is injective*

Show that if $f(x) = f(y)$ for arbitrary $x, y \in A$, then $x = y$.

- *To show that f is not injective*

Find particular elements $x, y \in A$ such that $x \neq y$ and $f(x) = f(y)$.

- *To show that f is surjective*

Consider an arbitrary element $y \in B$ and find an element $x \in A$ such that $f(x) = y$.

- *To show that f is not surjective*

Find a particular $y \in B$ such that $f(x) \neq y$ for all $x \in A$.

Showing that f is one-to-one or onto₂

Example 1: Let f be the function from $\{a,b,c,d\}$ to $\{1,2,3\}$ defined by $f(a) = 3$, $f(b) = 2$, $f(c) = 1$, and $f(d) = 3$. Is f an onto function? If the codomain were changed to $\{1,2,3,4\}$?

Solution: Yes, f is onto since all three elements of the codomain are images of elements in the domain.

If the codomain were changed to $\{1,2,3,4\}$, f would not be onto.

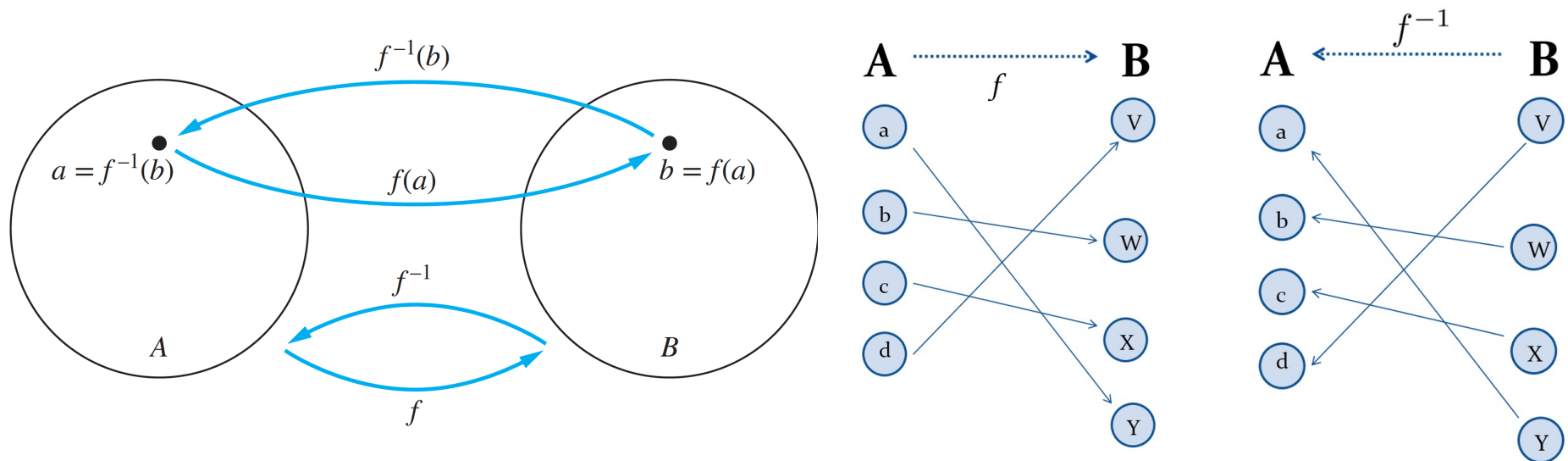
Example 2: Is the function $f(x) = x^2$ from the set of integers to the set of integers onto?

Solution: No, f is not onto because there is no integer x with $x^2 = -1$, for example.

2.3.3 Inverse Functions and Compositions of Functions

Inverse Functions₁

Definition: Let f be a bijection from A to B . Then the *inverse* of f , denoted f^{-1} is the function from B to A defined as $f^{-1}(y) = x$ if $f(x) = y$.



A one-to-one correspondence is called *invertible*.

A function is *not invertible* if it is not a bijection.

Questions₁

Example 1: Let f be the function from $\{a,b,c\}$ to $\{1,2,3\}$ such that $f(a) = 2$, $f(b) = 3$, and $f(c) = 1$. Is f invertible and if so what is its inverse?

Solution:

The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence given by f ,

so $f^{-1}(1) = c$, $f^{-1}(2) = a$, and $f^{-1}(3) = b$.

Questions₂

Example 2: Let $f: \mathbf{Z} \rightarrow \mathbf{Z}$ be such that $f(x) = x + 1$. Is f invertible, and if so, what is its inverse?

Solution:

The function f is invertible because it is a one-to-one correspondence. The inverse function f^{-1} reverses the correspondence so $f^{-1}(y) = y - 1$.

Questions₃

Example 3: Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be such that $f(x) = x^2$

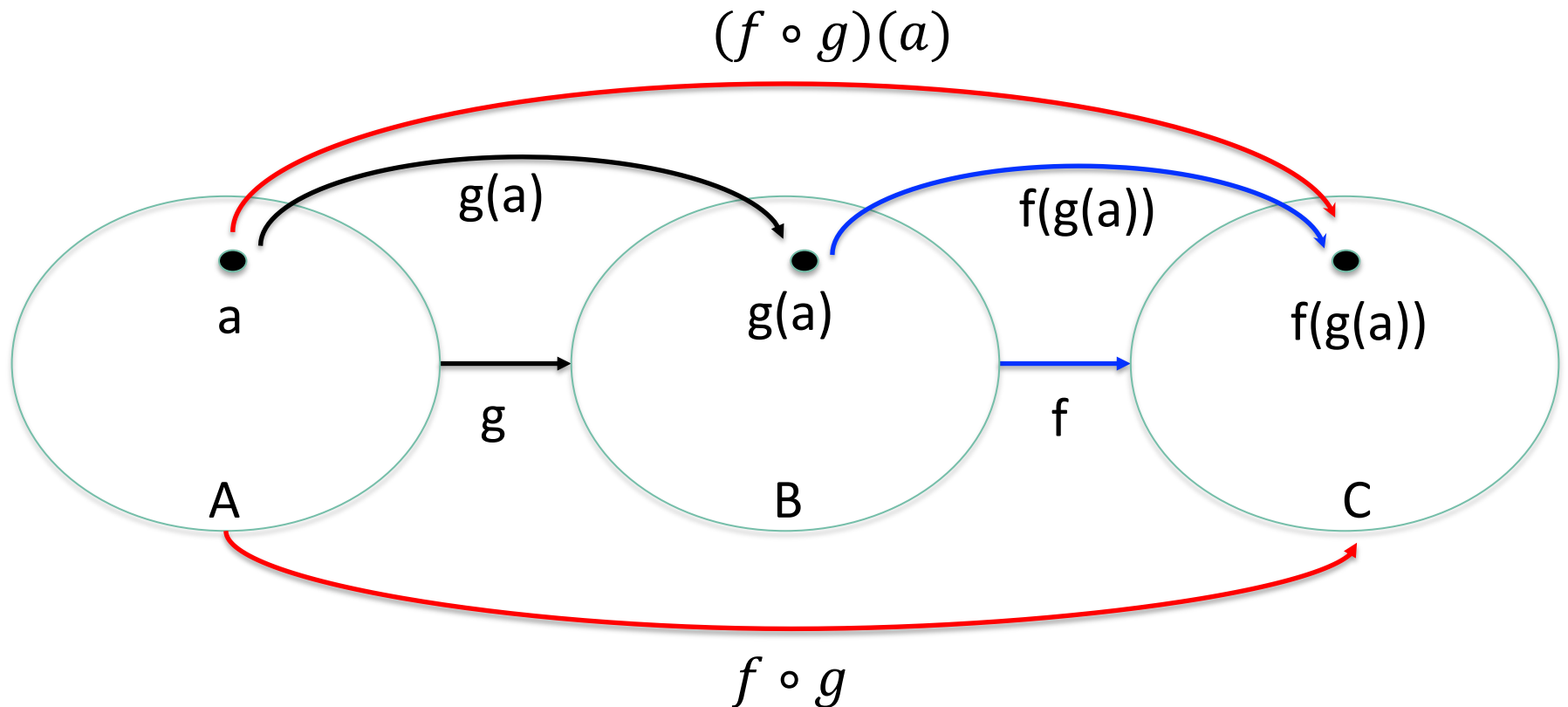
Is f invertible, and if so, what is its inverse?

Solution:

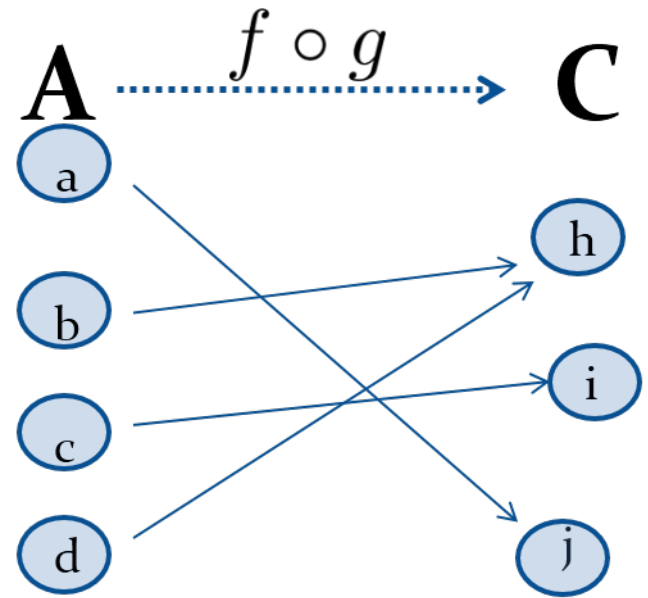
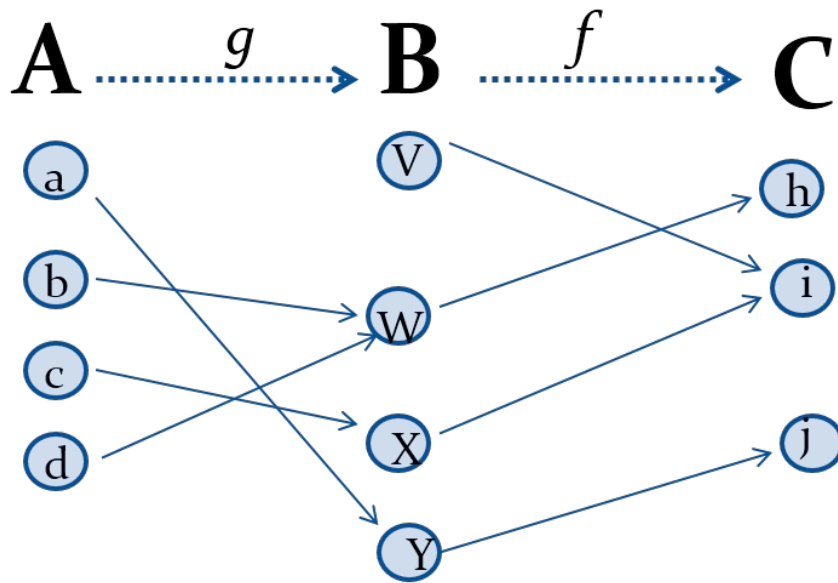
The function f is not invertible because it is not one-to-one.

Composition₁

Definition: Let $f: B \rightarrow C$, $g: A \rightarrow B$. The *composition of f with g* , denoted $f \circ g$ is the function from A to C defined by $f \circ g(x) = f(g(x))$



Composition₂



Composition₃

Example 1: If

$f(x) = x^2$ and $g(x) = 2x + 1$,
then

$$f(g(x)) = (2x + 1)^2$$

and

$$g(f(x)) = 2x^2 + 1$$

Composition Questions₁

Example 2: Let g be the function from the set $\{a,b,c\}$ to itself such that $g(a) = b$, $g(b) = c$, and $g(c) = a$. Let f be the function from the set $\{a,b,c\}$ to the set $\{1,2,3\}$ such that $f(a) = 3$, $f(b) = 2$, and $f(c) = 1$.

What is the composition of f and g , and what is the composition of g and f .

Solution: The composition $f \circ g$ is defined by

$$(f \circ g)(a) = f(g(a)) = f(b) = 2.$$

$$(f \circ g)(b) = f(g(b)) = f(c) = 1.$$

$$(f \circ g)(c) = f(g(c)) = f(a) = 3.$$

Note that $g \circ f$ is **not defined**, because the range of f is not a subset of the domain of g .

Composition Questions₂

Example 2: Let f and g be functions from the set of integers to the set of integers defined by

$$f(x) = 2x + 3 \text{ and } g(x) = 3x + 2.$$

What is the composition of f and g , and also the composition of g and f ?

Solution:

$$(f \circ g)(x) = f(g(x)) = f(3x + 2) = 2(3x + 2) + 3 = 6x + 7$$

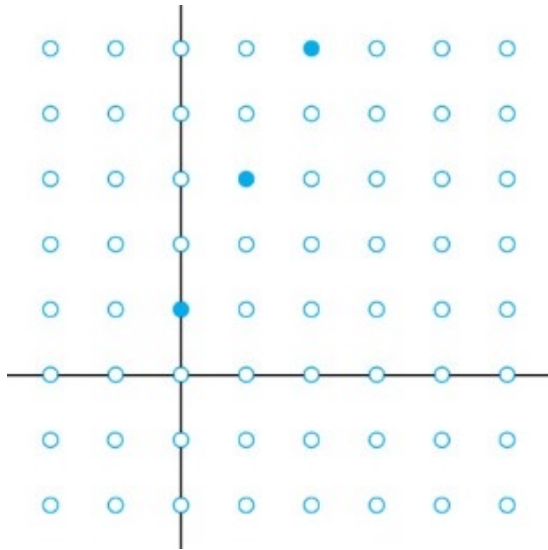
$$(g \circ f)(x) = g(f(x)) = g(2x + 3) = 3(2x + 3) + 2 = 6x + 11$$

2.3.4 Graphs of Functions

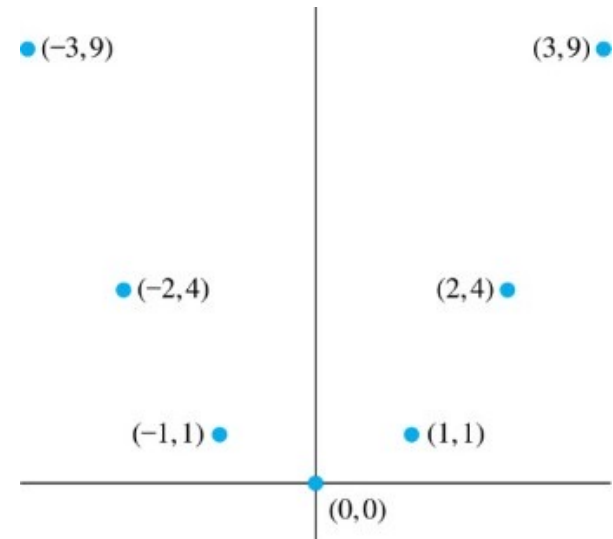
Let f be a function from the set A to the set B .

The *graph* of the function f is the set of ordered pairs

$$\{(a, b) | a \in A \text{ and } f(a) = b\}.$$



Graph of $f(n) = 2n + 1$
from $\mathbb{Z}$ to $\mathbb{Z}$



Graph of $f(x) = x^2$
from $\mathbb{Z}$ to $\mathbb{Z}$

2.3.5 Some Important Functions

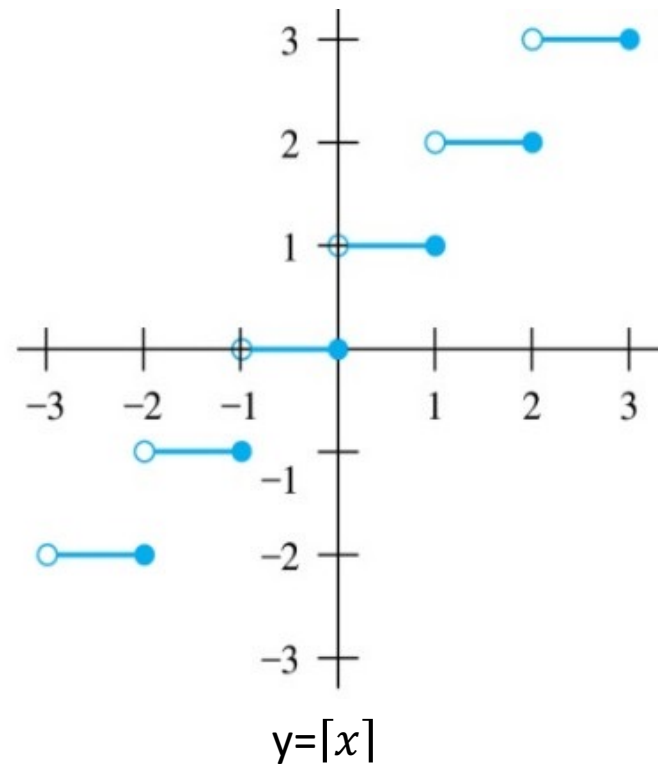
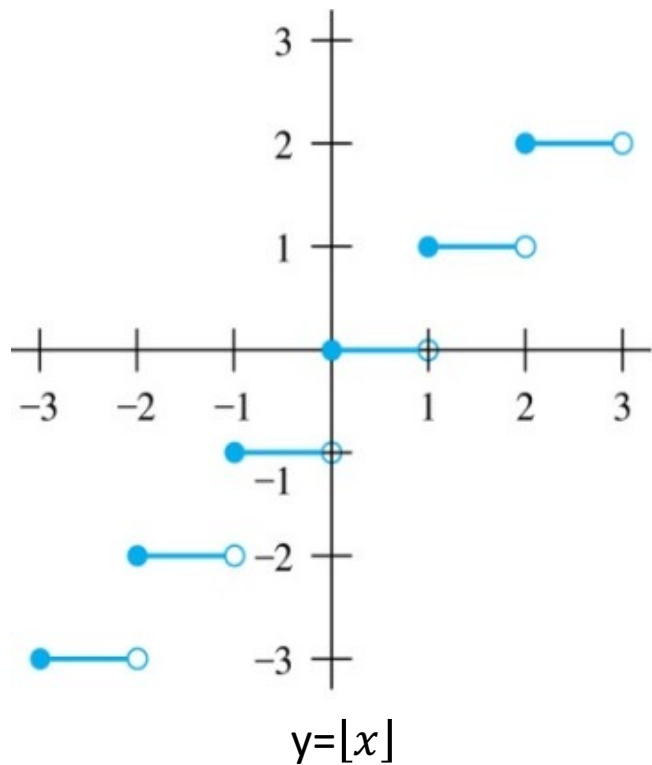
- The *floor* function, denoted $f(x) = \lfloor x \rfloor$ is the largest integer less than or equal to x .
- The *ceiling* function, denoted $f(x) = \lceil x \rceil$ is the smallest integer greater than or equal to x .

Example:

$$\lceil 3.5 \rceil = 4 \quad \lfloor 3.5 \rfloor = 3$$

$$\lceil -1.5 \rceil = -1 \quad \lfloor -1.5 \rfloor = -2$$

Floor and Ceiling Functions₁



Graph of (a) Floor and (b) Ceiling Functions

Floor and Ceiling Functions₂

TABLE 1 Useful Properties of the Floor and Ceiling Functions.

(n is an integer, x is a real number)

(1a) $\lfloor x \rfloor = n$ if and only if $n = x < n + 1$

(1b) $\lceil x \rceil = n$ if and only if $n - 1 < x = n$

(1c) $\lfloor x \rfloor = n$ if and only if $x - 1 < n = x$

(1d) $\lceil x \rceil = n$ if and only if $x = n < x + 1$

(2) $x - 1 < \lfloor x \rfloor \leq x \leq \lceil x \rceil < x + 1$

(3a) $\lfloor -x \rfloor = -\lceil x \rceil$

(3b) $\lceil -x \rceil = -\lfloor x \rfloor$

(4a) $\lfloor x + n \rfloor = \lfloor x \rfloor + n$

(4b) $\lceil x + n \rceil = \lceil x \rceil + n$

Proving Properties of Functions

Example: Prove that x is a real number, then

$$\lfloor 2x \rfloor = \lfloor x \rfloor + \lfloor x + 1/2 \rfloor$$

Solution: Let $x = n + \varepsilon$, where n is an integer and $0 \leq \varepsilon < 1$.

Case 1: $\varepsilon < 1/2$

- $2x = 2n + 2\varepsilon$ and $\lfloor 2x \rfloor = 2n$, since $0 \leq 2\varepsilon < 1$.
- $\lfloor x + 1/2 \rfloor = n$, since $x + 1/2 = n + (1/2 + \varepsilon)$ and $0 \leq 1/2 + \varepsilon < 1$.
- Hence, $\lfloor 2x \rfloor = 2n$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + n = 2n$.

Case 2: $\varepsilon \geq 1/2$

- $2x = 2n + 2\varepsilon = (2n + 1) + (2\varepsilon - 1)$ and $\lfloor 2x \rfloor = 2n + 1$, since $0 \leq 2\varepsilon - 1 < 1$.
- $\lfloor x + 1/2 \rfloor = \lfloor n + (1/2 + \varepsilon) \rfloor = \lfloor n + 1 + (\varepsilon - 1/2) \rfloor = n + 1$ since $0 \leq \varepsilon - 1/2 < 1$.
- Hence, $\lfloor 2x \rfloor = 2n + 1$ and $\lfloor x \rfloor + \lfloor x + 1/2 \rfloor = n + (n + 1) = 2n + 1$.

Factorial Function

Definition: $f: \mathbf{N} \rightarrow \mathbf{Z}^+$, denoted by $f(n) = n!$ is the product of the first n positive integers when n is a nonnegative integer.

$$f(n) = 1 \cdot 2 \cdots (n-1) \cdot n, f(0) = 0! = 1$$

$$f(1) = 1! = 1$$

$$f(2) = 2! = 1 \cdot 2 = 2$$

$$f(6) = 6! = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 = 720$$

$$f(20) = 2,432,902,008,176,640,000.$$

Stirling's Formula: $n! \sim \sqrt{2\pi n} (n/e)^n$

$$f(n) \sim g(n) \doteq \lim_{n \rightarrow \infty} f(n)/g(n) = 1$$

2.3.6 Partial Functions (*optional*)

Definition: A *partial function* f from a set A to a set B is an assignment to each element a in a subset of A , called the *domain of definition* of f , of a unique element b in B .

- The sets A and B are called the *domain* and *codomain* of f , respectively.
- We say that f is *undefined* for elements in A that are not in the domain of definition of f .
- When the domain of definition of f equals A , we say that f is a *total function*.

Example: $f: \mathbf{N} \rightarrow \mathbf{R}$ where $f(n) = \sqrt{n}$ is a partial function from $\mathbf{Z}$ to $\mathbf{R}$ where the domain of definition is the set of nonnegative integers. Note that f is undefined for negative integers.

Section Summary

2.3.1 Definition of a Function.

- Domain, Codomain
- Image, Preimage

2.3.2 Injection, Surjection, Bijection

2.3.3 Inverse Function and Compositions of Functions

2.3.4 Graphing Functions

2.3.5 Some Important Functions

Floor, Ceiling, Factorial

2.3.6 Partial Functions (optional)

Homework after class

2.3 Functions

- 8
- 23
- 69 (a and f)

Sequences and Summations

Section 2.4

2.4.2 Sequences₁

Sequences are ordered lists of elements.

- 1, 2, 3, 5, 8
- 1, 3, 9, 27, 81,

Definition: A **sequence** is a function from a subset of the integers (usually either the set $\{0, 1, 2, 3, 4, \dots\}$ or $\{1, 2, 3, 4, \dots\}$) to a set S .

The notation a_n is used to denote the image of the integer n . We can think of a_n as the equivalent of $f(n)$ where f is a function from $\{0, 1, 2, \dots\}$ to S . We call a_n a **term of the sequence**.

Sequences₂

Example: Consider the sequence $\{a_n\}$ where

$$a_n = \frac{1}{n}$$

$$a_1, a_2, a_3 \dots$$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$$

Geometric Progression

Definition: A *geometric progression* is a sequence of the form: $a, ar^2, \dots, ar^n, \dots$

where the *initial term* a and the *common ratio* r are real numbers.

Examples:

1. Let $a = 1$ and $r = -1$. Then:

$$\{b_n\} = \{b_0, b_1, b_2, b_3, b_4, \dots\} = \{1, -1, 1, -1, 1, \dots\}$$

2. Let $a = 2$ and $r = 5$. Then:

$$\begin{aligned}\{c_n\} &= \{c_0, c_1, c_2, c_3, c_4, \dots\} \\ &= \{2, 10, 50, 250, 1250, \dots\}\end{aligned}$$

3. Let $a = 6$ and $r = 1/3$. Then:

$$\{d_n\} = \{d_0, d_1, d_2, d_3, d_4, \dots\} = \left\{6, 2, \frac{2}{3}, \frac{2}{9}, \frac{2}{27}, \dots\right\}$$

Arithmetic Progression

Definition: A *arithmetic progression* is a sequence of the form: $a, a + d, a + 2d, \dots, a + nd, \dots$

where the *initial term* a and the *common difference* d are real numbers.

Examples :

1. Let $a = -1$ and $d = 4$:

$$\{s_n\} = \{s_0, s_1, s_2, s_3, s_4, \dots\} = \{-1, 3, 7, 11\}$$

2. Let $a = 7$ and $d = -3$:

$$\{t_n\} = \{t_0, t_1, t_2, t_3, t_4, \dots\} = \{7, 4, 1, -2, -5, \dots\}$$

3. Let $a = 1$ and $d = 2$:

$$\{u_n\} = \{u_0, u_1, u_2, u_3, u_4, \dots\} = \{1, 3, 5, 7, 9, \dots\}$$

Strings₁

Sequences of the form $a_1, a_2, \dots, a_n$ are often used in computer science. These finite sequences are also called **strings**.

This string is also denoted by $a_1a_2\dots a_n$.

The **length of a string** is the number of terms in this string.

The **empty string**, denoted by λ , is the string that has no terms.

The empty string has length zero.

The string *abcde* has *length* 5.

2.4.3 Recurrence Relations

Definition: A *recurrence relation* for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \dots, a_{n-1}$, for all integers n with $n \geq n_0$, where n_0 is a nonnegative integer.

A sequence is called a *solution* of a recurrence relation if its terms satisfy the recurrence relation.

Questions about Recurrence Relations₁

Example 1: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1, 2, 3, 4, \dots$ and suppose that $a_0 = 2$. What are a_1 , a_2 and a_3 ?

[Here $a_0 = 2$ is the initial condition.]

Solution: We see from the recurrence relation that

$$a_1 = a_0 + 3 = 2 + 3 = 5$$

$$a_2 = 5 + 3 = 8$$

$$a_3 = 8 + 3 = 11$$

Questions about Recurrence Relations₂

Example 2: Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} - a_{n-2}$ for $n = 2, 3, 4, \dots$ and suppose that $a_0 = 3$ and $a_1 = 5$. What are a_2 and a_3 ?

[Here the initial conditions are $a_0 = 3$ and $a_1 = 5$.]

Solution: We see from the recurrence relation that

$$\begin{aligned}a_2 &= a_1 - a_0 = 5 - 3 = 2 \\a_3 &= a_2 - a_1 = 2 - 5 = -3\end{aligned}$$

Fibonacci Sequence

Definition: Define the *Fibonacci sequence*, $f_0, f_1, f_2, \dots$, by:

- Initial Conditions: $f_0 = 0, f_1 = 1$
- Recurrence Relation: $f_n = f_{n-1} + f_{n-2}$

Example: Find f_2, f_3, f_4, f_5 and f_6 .

Answer:

$$f_2 = f_1 + f_0 = 1 + 0 = 1,$$

$$f_3 = f_2 + f_1 = 1 + 1 = 2,$$

$$f_4 = f_3 + f_2 = 2 + 1 = 3,$$

$$f_5 = f_4 + f_3 = 3 + 2 = 5,$$

$$f_6 = f_5 + f_4 = 5 + 3 = 8.$$

Solving Recurrence Relations

Finding a formula for the n th term of the sequence generated by a recurrence relation is called *solving the recurrence relation*. Such a formula is called a *closed formula* (e.g. $a_n = 2n + 3$).

Various methods for solving recurrence relations will be covered in Chapter 8 where recurrence relations will be studied in greater depth.

Here we illustrate by example the method of *iteration* in which we need to guess the formula. The guess can be proved correct by the method of *induction* (Chapter 5).

Iterative Solution Example₁

Method 1: Working upward, forward substitution Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

$$a_2 = 2 + 3$$

$$a_3 = (2 + 3) + 3 = 2 + 3 \cdot 2$$

$$a_4 = (2 + 2 \cdot 3) + 3 = 2 + 3 \cdot 3$$

.

.

.

$$a_n = a_{n-1} + 3 = (2 + 3 \cdot (n - 2)) + 3 = 2 + 3(n - 1)$$

Iterative Solution Example₂

Method 2: Working downward, backward substitution Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 2, 3, 4, \dots$ and suppose that $a_1 = 2$.

$$a_n = a_{n-1} + 3$$

$$= (a_{n-2} + 3) + 3 = a_{n-2} + 3 \cdot 2$$

$$= (a_{n-3} + 3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3$$

.

.

.

$$= a_2 + 3(n - 2) = (a_1 + 3) + 3(n - 2) = 2 + 3(n - 1)$$

Financial Application₁

Example: Suppose that a person deposits 10,000.00 in a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after n years?

Let P_n denote the amount in the account after n years. P_n satisfies the following recurrence relation:

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Financial Application₂

$$P_n = P_{n-1} + 0.11P_{n-1} = (1.11) P_{n-1}$$

with the initial condition $P_0 = 10,000$

Solution: Forward Substitution

$$P_1 = (1.11)P_0$$

$$P_2 = (1.11)P_1 = (1.11)^2 P_0$$

$$P_3 = (1.11)P_2 = (1.11)^3 P_0$$

:

$$P_n = (1.11)P_{n-1} = (1.11)^n P_0 = (1.11)^n 10,000$$

$P_n = (1.11)^n 10,000$ (Can prove by induction, covered in Chapter 5)

$$P_{30} = (1.11)^{30} 10,000 = 228,992.97$$

2.4.4 Special Integer Sequences (*opt*)

Given a few terms of a sequence, try to identify the sequence. Conjecture a formula, recurrence relation, or some other rule.

Some questions to ask?

- Are there **repeated** terms of the same value?
- Can you obtain a term from the previous term by **adding** an amount or multiplying by an amount?
- Can you obtain a term by **combining** the previous terms in some way?
- Are they **cycles** among the terms?
- Do the terms match those of a well **known** sequence?

Questions on Special Integer Sequences (*opt*)₁

Example 2: Consider 1,3,5,7,9

Solution: Note that each term is obtained by adding 2 to the previous term. A possible formula is $a_n = 2n + 1$. This is an arithmetic progression with $a = 1$ and $d = 2$.

Example 3: 1, -1, 1, -1,1

Solution: The terms alternate between 1 and -1. A possible sequence is $a_n = (-1)^n$. This is a geometric progression with $a = 1$ and $r = -1$.

Example 1: Find formulae for the sequences with the following first five terms: 1, 1/2, 1/4, 1/8, 1/16

Solution: Note that the denominators are powers of 2. The sequence with $a_n = 1/2^n$ is a possible match. This is a geometric progression with $a = 1$ and $r = 1/2$.

Questions on Special Integer Sequences (*opt*)₂

TABLE 1 Some Useful Sequences.

<i>nth Term</i>	<i>First 10 Terms</i>
n^2	1, 4, 9, 16, 25, 36, 49, 64, 81, 100,...
n^3	1, 8, 27, 64, 125, 216, 343, 512, 729, 1000,...
n^4	1, 16, 81, 256, 625, 1296, 2401, 4096, 6561, 10000,...
f_n	1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89,...
2^n	2, 4, 8, 16, 32, 64, 128, 256, 512, 1024,...
3^n	3, 9, 27, 81, 243, 729, 2187, 6561, 19683, 59049,...
$n!$	1, 2, 6, 24, 120, 720, 5040, 40320, 362880, 3628800,...

Guessing Sequences (*optional*)

Example: Conjecture a simple formula for a_n if the first 10 terms of the sequence $\{a_n\}$ are 1, 7, 25, 79, 241, 727, 2185, 6559, 19681, 59047.

Solution:

Note the ratio of each term to the previous approximates 3.

So now compare with the sequence 3^n . We notice that the n th term is 2 less than the corresponding power of 3.

So a good conjecture is that $a_n = 3^n - 2$.

2.4.5 Summations₁

Sum of the terms $a_m, a_m + 1, \dots, a_n$
from the sequence $\{a_n\}$

Σ
Sigma

The notation:

$$\sum_{j=m}^n a_j \quad \text{or} \quad \sum_{j=m}^n a_j \quad \text{or} \quad \sum_{m \leq j \leq n} a_j$$

represents

$$a_m + a_{m+1} + \dots + a_n$$

The variable j is called the *index of summation*. It runs through all the integers starting with *its lower limit* m and ending with its *upper limit* n .

Summations₂

More generally for a set S :

$$\sum_{j \in S} a_j$$

Examples:

$$r^0 + r^1 + r^2 + r^3 + \dots + r^n = \sum_0^n r^j$$

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots = \sum_1^{\infty} \frac{1}{i}$$

$$\text{If } S = \{2, 5, 7, 10\} \text{ then } \sum_{j \in S} a_j = a_2 + a_5 + a_7 + a_{10}$$

Product Notation (supplement)

Product of the terms $a_m, a_m + 1, \dots, a_n$
from the sequence $\{a_n\}$

π

The notation:

$$\prod_{j=m}^n a_j \quad \text{or} \quad \prod_{j=m}^n a_j \quad \text{or} \quad \prod_{m \leq j \leq n} a_j$$

represents

$$a_m \times a_{m+1} \times \cdots \times a_n$$

Geometric Series₁

Sums of terms of geometric progressions

$$\sum_{j=0}^n ar^j = \begin{cases} \frac{ar^{n+1} - a}{r - 1} & r \neq 1 \\ (n + 1)a & r = 1 \end{cases}$$

Proof: Let

$$S_n = \sum_{j=0}^n ar^j$$

$$rS_n = r \sum_{j=0}^n ar^j$$

$$= \sum_{j=0}^n ar^{j+1}$$

To compute S_n , first multiply both sides of the equality by r and then manipulate the resulting sum as follows:

Geometric Series₂

$$= \sum_{j=0}^n ar^{j+1}$$

From previous slide.

$$= \sum_{k=1}^{n+1} ar^k$$

Shifting the index of summation with $k = j + 1$.

$$= \left(\sum_{k=0}^n ar^k \right) + (ar^{n+1} - a)$$

Removing $k = n + 1$ term and adding $k = 0$ term.

$$= S_n + (ar^{n+1} - a)$$

Substituting S for summation formula

$$\therefore rS_n = S_n + (ar^{n+1} - a)$$

$$S_n = \frac{ar^{n+1} - a}{r - 1}$$

if $r \neq 1$

$$S_n = \sum_{j=0}^n ar^j = \sum_{j=0}^n a = (n + 1)a$$

if $r = 1$

Some Useful Summation Formulae

TABLE 2 Some Useful Summation Formulae.

<i>Sum</i>	<i>Closed Form</i>
$\sum_{k=0}^n ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=0}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

Geometric Series:
We just proved this.

Later we will
prove some
of these by
induction.

Proof in text
(requires calculus)

Section Summary

Sequences.

- Geometric Progression, Arithmetic Progression

Recurrence Relations

- Fibonacci Sequence

Summations

Special Integer Sequences (*optional*)

Homework after class

2.4 Sequences and Summations

- 3
- 29

Cardinality of Sets

Section 2.5

2.5.1 Cardinality

Definition: The *cardinality* of a set A is equal to the cardinality of a set B , denoted

$$|A| = |B|,$$

if and only if there is a one-to-one correspondence (*i.e.*, a bijection) from A to B .

If there is a one-to-one function (*i.e.*, an injection) from A to B , the cardinality of A is less than or the same as the cardinality of B and we write $|A| \leq |B|$.

When $|A| \leq |B|$ and A and B have different cardinality, we say that the cardinality of A is less than the cardinality of B and write $|A| < |B|$.

2.5.2 Countable Sets

Definition: A set that is either finite or has the same cardinality as the set of positive integers ($\mathbf{Z}^+$) is called *countable*. A set that is not countable is *uncountable*.

The set of real numbers $\mathbf{R}$ is an uncountable set.

When an infinite set is countable (*countably infinite*) its cardinality is $\aleph_0$ (where $\aleph$ is aleph, the 1st letter of the *Hebrew alphabet*). We write $|S| = \aleph_0$ and say that S has cardinality “*aleph null*.”

Showing that a Set is Countable₁

An infinite set is **countable** if and only if it is possible to list the elements of the set in a sequence (indexed by the positive integers).

The reason for this is that a **one-to-one correspondence** f from the **set of positive integers** to a set S can be expressed in terms of a sequence $a_1, a_2, \dots, a_n, \dots$ where $a_1 = f(1)$, $a_2 = f(2)$, $\dots$, $a_n = f(n)$, $\dots$

Showing that a Set is Countable₂

Show that the set of positive even integers E is countable set.

To show that a set is countable, we will exhibit a one-to-one correspondence between that set and the set of positive integers.

Solution:

$$f(x) = 2x.$$

1	2	3	4	5	6
↕	↕	↕	↕	↕	↕
2	4	6	8	10	12

Then f is a bijection from $\mathbf{N}$ to E since f is both one-to-one and onto.

To show that it is one-to-one, suppose that $f(n) = f(m)$. Then $2n = 2m$, and so $n = m$.

To see that it is onto, suppose that t is an even positive integer. Then $t = 2k$ for some positive integer k and $f(k) = t$.

Showing that a Set is Countable₃

Show that the set of integers **Z** is countable.

Solution: Can list in a **sequence**:

0, 1, - 1, 2, - 2, 3, - 3 ,.....

Or can define a **bijection** from **N** to **Z**:

- When n is even: $f(n) = n/2$
- When n is odd: $f(n) = -(n-1)/2$

The Positive Rational Numbers are Countable₁

Definition: A *rational number* can be expressed as the ratio of two integers p and q such that $q \neq 0$.

- $3/4$ is a rational number
- $\sqrt{2}$ is not a rational number.

Show that the positive rational numbers are countable.

Solution: The positive rational numbers are countable since they can be arranged in a sequence:

$$r_1, r_2, r_3, \dots$$

The next slide shows how this is done.

The Positive Rational Numbers are Countable₂

Constructing the List

First list p/q with $p + q = 2$.

Next list p/q with $p + q = 3$

And so on.

First row $q = 1$.

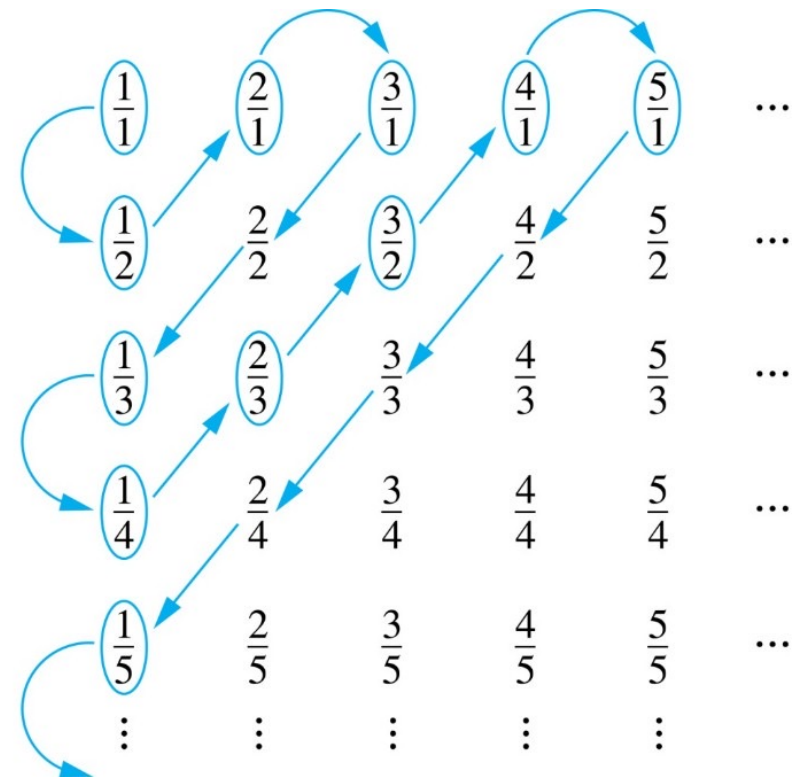
Second row $q = 2$.

etc.

Terms not circled
are not listed
because they
repeat previously
listed terms

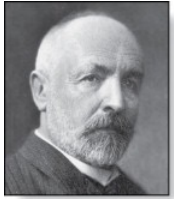
1, 1/2, 2, 3, 1/3, 1/4,

2/3, 3/2, 4, 5,



The Real Numbers are Uncountable

Georg Cantor
(1845-1918)



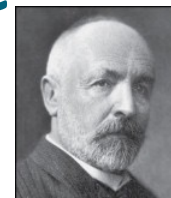
Example: Show that the set of **real numbers** is **uncountable**.

Solution: The method is called the Cantor diagonalization argument, and is a proof by contradiction.

1. Suppose **$\mathbf{R}$ is countable**. Then the real numbers between 0 and 1 are also countable (any subset of a countable set is countable - an exercise in the text).
2. The real numbers between **0 and 1 can be listed** in order $r_1, r_2, r_3, \dots$.
3. Let the decimal representation of this listing be $r_1 = 0.d_{11}d_{12}d_{13}d_{14}d_{15}d_{16}\dots$
4. Form a **new real number** with the decimal expansion $r = .r_1r_2r_3r_4\dots$
where $r_i = 3$ if $d_{ii} \neq 3$ and $r_i = 4$ if $d_{ii} = 3$
 $r_2 = 0.d_{21}d_{22}d_{23}d_{24}d_{25}d_{26}\dots$
5. r is **not equal** to any of the $r_1, r_2, r_3, \dots$. **Because it differs from r_i in its i th position** after the decimal point. Therefore there is a real number between 0 and 1 that is **not on the list** since every real number has a unique decimal expansion. Hence, all the real numbers between 0 and 1 **cannot be listed**, so the set of real numbers between 0 and 1 is **uncountable**.
6. Since a set with an uncountable subset is uncountable (an exercise), the set of real numbers is uncountable.

The Real Numbers are Uncountable

Georg Cantor
(1845-1918)



Example: Show that the set of **real numbers** is **uncountable**.

Solution: The method is called the Cantor diagonalization argument, and is a proof by contradiction.

$$r_1 = 0.d_{11}d_{12}d_{13}d_{14} \dots$$

$$r_2 = 0.d_{21}d_{22}d_{23}d_{24} \dots$$

$$r_3 = 0.d_{31}d_{32}d_{33}d_{34} \dots$$

$$r_4 = 0.d_{41}d_{42}d_{43}d_{44} \dots$$

$\vdots$

where $d_{ij} \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Form a **new real number** with the decimal expansion:

$$d_i = \begin{cases} 4 & \text{if } d_{ii} \neq 4 \\ 5 & \text{if } d_{ii} = 4. \end{cases}$$

$$r = 0.d_1 d_2 d_3 d_4 \dots,$$

$$r = 0.4544\dots,$$

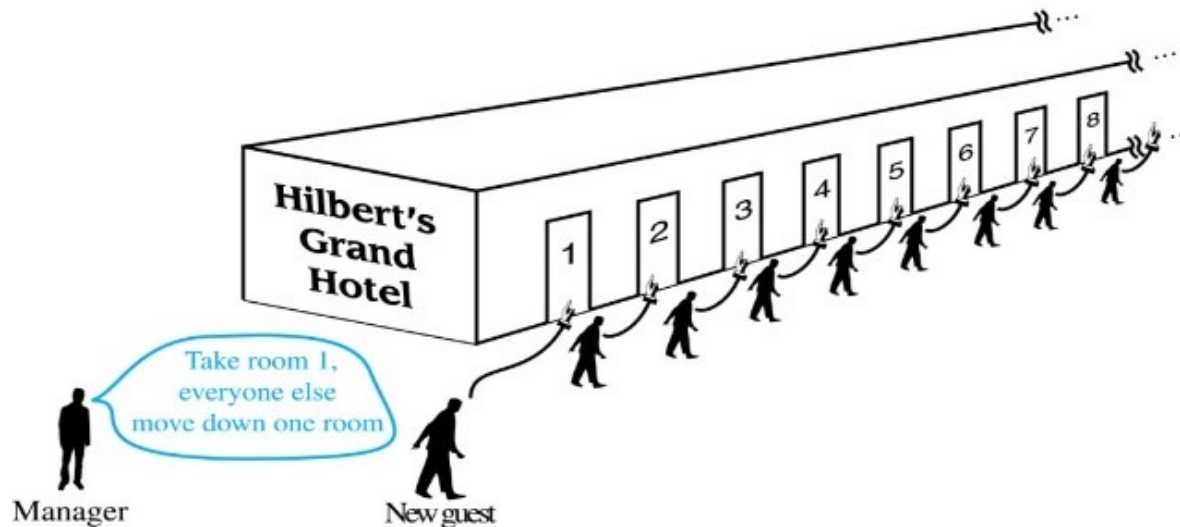
$$r \neq r_1, \neq r_2, \neq r_3, \neq r_4, \neq \dots$$

Hilbert's Grand Hotel

The **Grand Hotel** (example due to David Hilbert) has **countably infinite number of rooms**, each occupied by a guest. We can always accommodate a new guest at this hotel. How is this possible?



David Hilbert



Explanation: Because the rooms of Grand Hotel are countable, we can list them as Room 1, Room 2, Room 3, and so on. When a new guest arrives, we move the guest in Room 1 to Room 2, the guest in Room 2 to Room 3, and in general the guest in Room n to Room $n + 1$, for all positive integers n . This frees up Room 1, which we assign to the new guest, and all the current guests still have rooms.

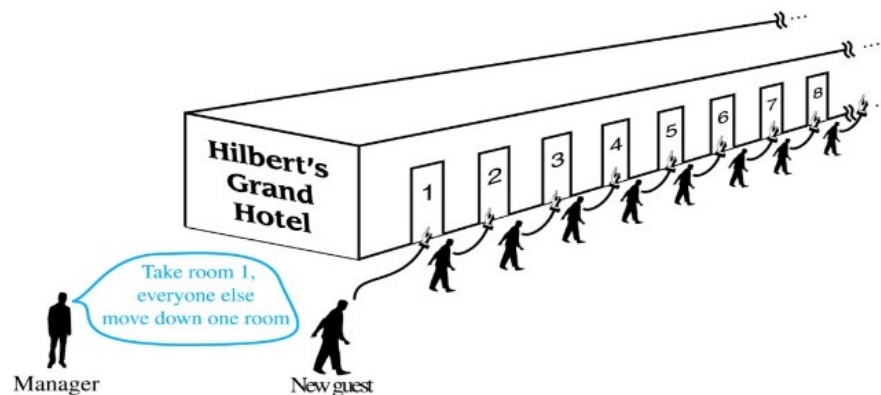
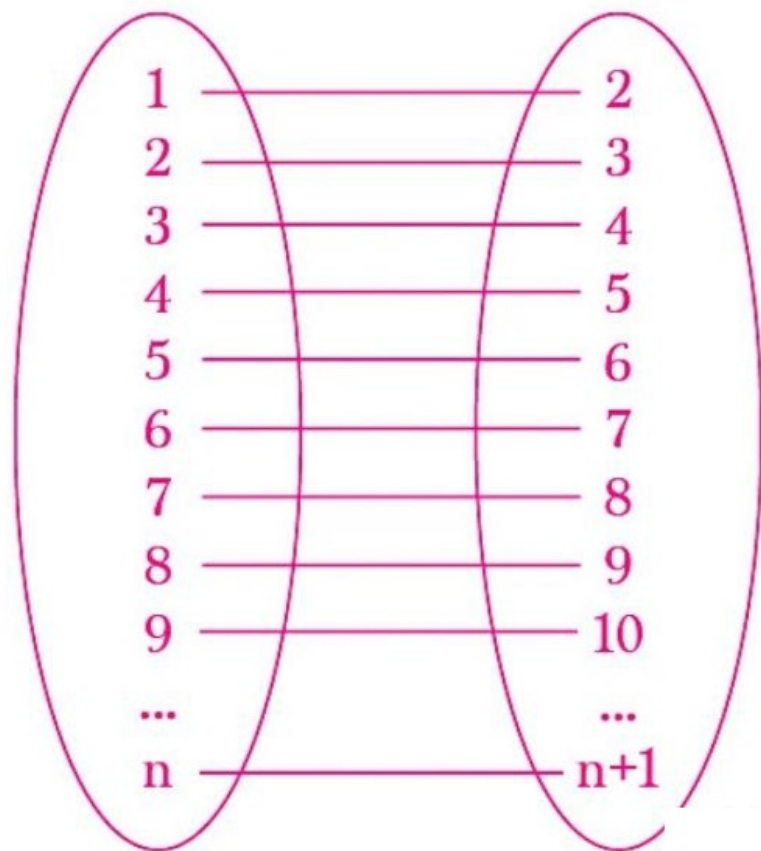
Hilbert's Grand Hotel



David Hilbert

The
Infinite Hotel's
Rooms

The
Infinite Hotel's
Rooms plus 1



Results about cardinality

If A and B are countable sets, then $A \cup B$ is also countable.

Case (i): A and B are both finite

$A \cup B$ is also finite, and therefore, countable.

Case (ii): A is infinite and B is finite (or A and B exchange)

A : $a_1, a_2, a_3, \dots, a_n, \dots$

B : $b_1, b_2, \dots, b_m$.

$A \cup B$ as $b_1, b_2, \dots, b_m, a_1, a_2, a_3, \dots, a_n, \dots$

$A \cup B$ must be countably infinite.

Case (iii): A and B are both countably infinite.

By alternating terms of these two sequences : $a_1, b_1, a_2, b_2, a_3, b_3, \dots, a_n, b_n, \dots$

$A \cup B$ must be countably infinite.

Results about cardinality

Schroder-Bernstein theorem:

If A and B are sets with $|A| \leq |B|$ and $|B| \leq |A|$, then $|A| = |B|$.
In other words, if there are one-to-one functions f from A to B and g from B to A , then there is a one-to-one correspondence between A and B .

Show that the $|(0, 1)| = |(0, 1]|$.

$f(x) = x$ is a one-to-one function from $(0, 1)$ to $(0, 1]$.

$g(x) = x/2$ is a one-to-one function from $(0, 1]$ to $(0, 1/2] \subset (0, 1)$.

The Continuum hypothesis (Optional)

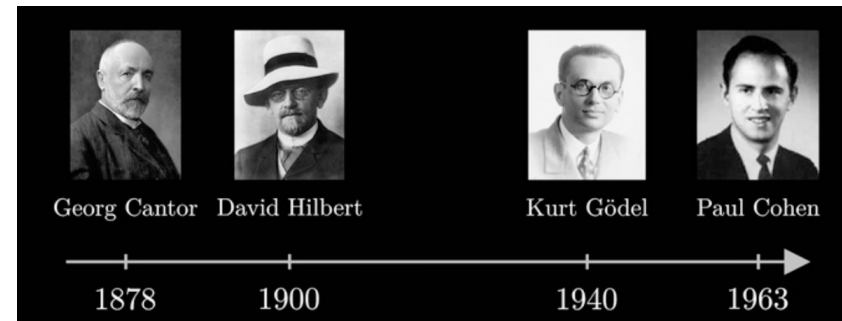
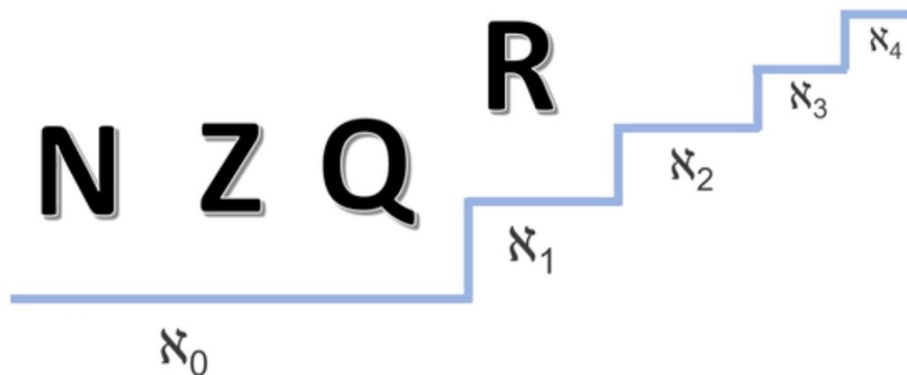
$$\underbrace{\mathbb{N}^* \sim 2\mathbb{N}^* \sim \mathbb{Z} \sim \mathbb{N} \times \mathbb{N} \sim \mathbb{Q} \sim \dots}_{\aleph_0}$$

$$\underbrace{(0, 1) \sim [0, 1] \sim \mathbb{R} \sim \mathbb{C} \sim \dots}_{\aleph_1}$$

$$\aleph_0 < \aleph_1 < \aleph_2 < \aleph_3 < \dots$$



There is no intermediate cardinality



Section Summary

Cardinality

Countable Sets

Computability (Optional)

Extra materials

https://www.bilibili.com/video/BV1op4y1q7vR/?spm_id_from=333.788.recommend_more_video.4

Homework after class

2.5 Cardinality of Sets

- 1
- 10

Matrices (Options)

Section 2.6

Matrices

Matrices are used throughout discrete mathematics **to express relationships between elements in sets**.

Matrices are useful discrete structures that can be used in many ways. For example, they are used to:

- Describe certain types of functions known as linear transformations.
- Express which vertices of a graph are connected by edges (Ch. 10).

In later chapters, we will see matrices **used to build models** of:

- Transportation systems.
- Communication networks.

Algorithms based on matrix models will be presented in later chapters.

Here we cover the aspect of **matrix arithmetic** that will be needed later.

Matrix

Definition: A *matrix* is a rectangular array of numbers. A matrix with m rows and n columns is called an $m \times n$ matrix.

- The plural of matrix is *matrices*.
- A matrix with the same number of rows as columns is called *square*.
- Two matrices are *equal* if they have the same number of rows and the same number of columns and the corresponding entries in every position are equal.

$$\begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 1 & 3 \end{bmatrix}$$

3×2 matrix

Notation

Let m and n be positive integers and let $\mathbf{A}$

$$= \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

The i th row of $\mathbf{A}$ is the $1 \times n$ matrix $[a_{i1}, a_{i2}, \dots, a_{in}]$. The j th column of $\mathbf{A}$ is the $m \times 1$ matrix:

$$\begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix}$$

The (i,j) th *element* or *entry* of $\mathbf{A}$ is the element a_{ij} . We can use $\mathbf{A} = [a_{ij}]$ to denote the matrix with its (i,j) th element equal to a_{ij} .

Matrix Arithmetic: Addition

Definition: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be $m \times n$ matrices. The sum of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} + \mathbf{B}$, is the $m \times n$ matrix that has $a_{ij} + b_{ij}$ as its (i,j) th element. In other words, $\mathbf{A} + \mathbf{B} = [a_{ij} + b_{ij}]$.

Example:

$$\begin{bmatrix} 1 & 0 & -1 \\ 2 & 2 & -3 \\ 3 & 4 & 0 \end{bmatrix} + \begin{bmatrix} 3 & 4 & -1 \\ 1 & -3 & 0 \\ -1 & 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 4 & -2 \\ 3 & -1 & -3 \\ 2 & 5 & 2 \end{bmatrix}$$

Note that matrices of different sizes **can not** be added.

Matrix Multiplication

Definition: Let $\mathbf{A}$ be an $m \times k$ matrix and $\mathbf{B}$ be a $k \times n$ matrix. The *product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{AB}$, is the $m \times n$ matrix that has its (i,j) th element equal to the sum of the products of the corresponding elements from *the i th row of $\mathbf{A}$ and the j th column of $\mathbf{B}$* . In other words, if $\mathbf{AB} = [c_{ij}]$ then $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{kj}b_{kj}$.

Example:

$$\begin{bmatrix} 1 & 0 & 4 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \\ 0 & 2 & 2 \end{bmatrix} \begin{bmatrix} 2 & 4 \\ 1 & 1 \\ 3 & 0 \end{bmatrix} = \begin{bmatrix} 14 & 4 \\ 8 & 9 \\ 7 & 13 \\ 8 & 2 \end{bmatrix}$$

The product of two matrices is undefined when the number of *columns* in the first matrix is *not the same* as the number of *rows* in the second.

Illustration of Matrix Multiplication

The Product of $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1k} \\ a_{21} & a_{22} & \dots & a_{2k} \\ \vdots & \vdots & & \vdots \\ \color{red}{a_{i1}} & \color{red}{a_{i2}} & \dots & \color{red}{a_{ik}} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mk} \end{bmatrix}$$

$$\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & \dots & \color{red}{b_{1j}} & \dots & b_{1n} \\ b_{21} & b_{22} & \dots & \color{red}{b_{2j}} & \dots & b_{2n} \\ \vdots & \vdots & & \vdots & & \vdots \\ \vdots & \vdots & & \vdots & & \vdots \\ b_{k1} & b_{k2} & \dots & \color{red}{b_{kj}} & \dots & b_{kn} \end{bmatrix}$$

$$\mathbf{AB} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1n} \\ c_{21} & c_{22} & \dots & c_{2n} \\ \vdots & \vdots & & \vdots \\ \vdots & \vdots & \color{red}{c_{ij}} & \vdots \\ \vdots & \vdots & & \vdots \\ c_{m1} & c_{m2} & \dots & c_{mn} \end{bmatrix}$$

$$\color{red}{c_{ij}} = a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ik}b_{kj}$$

Matrix Multiplication is not Commutative

Example: Let

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$$

Does **$AB = BA$** ?

Solution:

$$AB = \begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix}$$

$\neq$

$$BA = \begin{bmatrix} 4 & 3 \\ 3 & 2 \end{bmatrix}$$

Identity Matrix and Powers of Matrices

Definition: The *identity matrix* of order n is the $m \times n$ matrix $\mathbf{I}_n = [\delta_{ij}]$, where $\delta_{ij} = 1$ if $i = j$ and $\delta_{ij} = 0$ if $i \neq j$.

$$\mathbf{I}_n = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix}$$

$$\mathbf{A}\mathbf{I}_n = \mathbf{I}_m\mathbf{A} = \mathbf{A}$$

when $\mathbf{A}$ is an $m \times n$ matrix

Powers of square matrices can be defined. When $\mathbf{A}$ is an $n \times n$ matrix, we have:

$$\mathbf{A}_0 = \mathbf{I}_n \quad \mathbf{A}^r = \underbrace{\mathbf{A}\mathbf{A}\mathbf{A} \cdots \mathbf{A}}_{r \text{ times}}$$

Transposes of Matrices₁

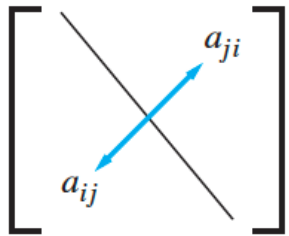
Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times n$ matrix. The *transpose* of $\mathbf{A}$, denoted by $\mathbf{A}^t$, is the $n \times m$ matrix obtained by interchanging the rows and columns of $\mathbf{A}$.

If $\mathbf{A}^t = [b_{ij}]$, then $b_{ij} = a_{ji}$ for $i = 1, 2, \dots, n$ and $j = 1, 2, \dots, m$.

The transpose of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ is the matrix $\begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$

Transposes of Matrices₂

Definition: A square matrix $\mathbf{A}$ is called **symmetric** if $\mathbf{A} = \mathbf{A}^t$. Thus $\mathbf{A} = [a_{ij}]$ is symmetric if $a_{ij} = a_{ji}$ for i and j with $1 \leq i \leq n$ and $1 \leq j \leq n$.



The matrix $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ is square.

Square matrices do not change when their rows and columns are interchanged.

Zero-One Matrices₁

Definition: A matrix all of whose entries are either 0 or 1 is called a *zero-one matrix*. (These will be used in Chapters 9 and 10.)

Algorithms operating on discrete structures represented by zero-one matrices are based on Boolean arithmetic defined by the following Boolean operations:

$$b_1 \wedge b_2 = \begin{cases} 1 & \text{if } b_1 = b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

$$b_1 \vee b_2 = \begin{cases} 1 & \text{if } b_1 = 1 \text{ or } b_2 = 1 \\ 0 & \text{otherwise} \end{cases}$$

Zero-One Matrices₂

Definition: Let $\mathbf{A} = [a_{ij}]$ and $\mathbf{B} = [b_{ij}]$ be an $m \times n$ zero-one matrices.

- The *join* of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \vee b_{ij}$. The *join* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \vee \mathbf{B}$.
- The *meet* of $\mathbf{A}$ and $\mathbf{B}$ is the zero-one matrix with (i,j) th entry $a_{ij} \wedge b_{ij}$. The *meet* of $\mathbf{A}$ and $\mathbf{B}$ is denoted by $\mathbf{A} \wedge \mathbf{B}$.

Joins and Meets of Zero-One Matrices

Example: Find the join and meet of the zero-one matrices

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix},$$

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Solution: The **join** of **A** and **B** is

$$A \vee B = \begin{bmatrix} 1 & \vee & 0 & 0 & \vee & 1 & 1 & \vee & 0 \\ 0 & \vee & 1 & 1 & \vee & 1 & 0 & \vee & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

The **meet** of **A** and **B** is

$$A \wedge B = \begin{bmatrix} 1 & \wedge & 0 & 0 & \wedge & 1 & 1 & \wedge & 0 \\ 0 & \wedge & 1 & 1 & \wedge & 1 & 0 & \wedge & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Boolean Product of Zero-One Matrices₁

Definition: Let $\mathbf{A} = [a_{ij}]$ be an $m \times k$ zero-one matrix and $\mathbf{B} = [b_{ij}]$ be a $k \times n$ zero-one matrix. The *Boolean product* of $\mathbf{A}$ and $\mathbf{B}$, denoted by $\mathbf{A} \odot \mathbf{B}$, is the $m \times n$ zero-one matrix with (i,j) th entry

$$c_{ij} = (a_{i1} \wedge b_{1j}) \vee (a_{i2} \wedge b_{2j}) \vee \dots \vee (a_{ik} \wedge b_{kj}).$$

Note that the Boolean product of $\mathbf{A}$ and $\mathbf{B}$ is obtained in an analogous way to the ordinary product of these matrices, but with *addition* replaced with the operation $\vee$ and with *multiplication* replaced with the operation $\wedge$.

Example: Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Boolean Product of Zero-One Matrices₂

Solution: The Boolean product $\mathbf{A} \odot \mathbf{B}$ is given by

$$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0)(1 \wedge 1) \vee (0 \wedge 1)(1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0)(0 \wedge 1) \vee (1 \wedge 1)(0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0)(1 \wedge 1) \vee (0 \wedge 1)(1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \vee & 01 & \vee & 00 & \vee & 0 \\ 0 & \vee & 00 & \vee & 10 & \vee & 1 \\ 1 & \vee & 01 & \vee & 00 & \vee & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Boolean Powers of Zero-One Matrices₁

Definition: Let $\mathbf{A}$ be a square zero-one matrix and let r be a positive integer. The r th Boolean power of $\mathbf{A}$ is the Boolean product of r factors of $\mathbf{A}$, denoted by $\mathbf{A}^{[r]}$. Hence,

$$\mathbf{A}^{[r]} = \underbrace{\mathbf{A} \odot \mathbf{A} \odot \cdots \odot \mathbf{A}}_{r \text{ times}}.$$

(The Boolean product is well defined because the Boolean product of matrices is associative.)

Boolean Powers of Zero-One Matrices₂

Example: Let

$$A = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix}.$$

Find A^n for all positive integers n .

Solution:

$$\begin{aligned} A^{[2]} &= A \odot A \\ &= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A^{[3]} &= A^{[2]} \odot A \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} A^{[4]} &= A^{[3]} \odot A \\ &= \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \end{aligned}$$

$$A^{[5]} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad A^{[n]} = A^5 \text{ for all positive integers } n \text{ with } n \geq 5$$

Section Summary

Definition of a Matrix

Matrix Arithmetic

Transposes and Powers of Arithmetic

Zero-One matrices

Chapter Summary

Sets

- The Language of Sets
- Set Operations
- Set Identities

Functions

- Types of Functions
- Operations on Functions
- Computability

Sequences and Summations

- Types of Sequences
- Summation Formulae

Set Cardinality

- Countable Sets

Matrices

- Matrix Arithmetic

Key Terms and Results

TERMS

set: an unordered collection of distinct objects

axiom: a basic assumption of a theory

paradox: a logical inconsistency

element, member of a set: an object in a set

roster method: a method that describes a set by listing its elements

set builder notation: the notation that describes a set by stating a property an element must have to be a member

multiset: an unordered collection of objects where objects can occur multiple times

$\emptyset$ (empty set, null set): the set with no members

universal set: the set containing all objects under consideration

Venn diagram: a graphical representation of a set or sets

$S = T$ (set equality): S and T have the same elements

$S \subseteq T$ (S is a subset of T): every element of S is also an element of T

$S \subset T$ (S is a proper subset of T): S is a subset of T and $S \neq T$

finite set: a set with n elements, where n is a nonnegative integer

infinite set: a set that is not finite

$|S|$ (the cardinality of S): the number of elements in S

$P(S)$ (the power set of S): the set of all subsets of S

$A \cup B$ (the union of A and B): the set containing those elements that are in at least one of A and B

$A \cap B$ (the intersection of A and B): the set containing those elements that are in both A and B .

$A - B$ (the difference of A and B): the set containing those elements that are in A but not in B

$\overline{A}$ (the complement of A): the set of elements in the universal set that are not in A

$A \oplus B$ (the symmetric difference of A and B): the set containing those elements in exactly one of A and B

membership table: a table displaying the membership of elements in sets

function from A to B : an assignment of exactly one element of B to each element of A

domain of f : the set A , where f is a function from A to B

codomain of f : the set B , where f is a function from A to B

b is the image of a under f : $b = f(a)$

a is a preimage of b under f : $f(a) = b$

range of f : the set of images of f

onto function, surjection: a function from A to B such that every element of B is the image of some element in A

one-to-one function, injection: a function such that the images of elements in its domain are distinct

one-to-one correspondence, bijection: a function that is both one-to-one and onto

inverse of f : the function that reverses the correspondence given by f (when f is a bijection)

$f \circ g$ (composition of f and g): the function that assigns $f(g(x))$ to x

$\lfloor x \rfloor$ (floor function): the largest integer not exceeding x

$\lceil x \rceil$ (ceiling function): the smallest integer greater than or equal to x

partial function: an assignment to each element in a subset of the domain a unique element in the codomain

sequence: a function with domain that is a subset of the set of integers

geometric progression: a sequence of the form $a, ar, ar^2, \dots$, where a and r are real numbers

arithmetic progression: a sequence of the form $a, a + d, a + 2d, \dots$, where a and d are real numbers

string: a finite sequence

empty string: a string of length zero

recurrence relation: a equation that expresses the n th term a_n of a sequence in terms of one or more of the previous terms of the sequence for all integers n greater than a particular integer

$\sum_{i=1}^n a_i$: the sum $a_1 + a_2 + \dots + a_n$

$\prod_{i=1}^n a_i$: the product $a_1 a_2 \dots a_n$

cardinality: two sets A and B have the same cardinality if there is a one-to-one correspondence from A to B

countable set: a set that either is finite or can be placed in one-to-one correspondence with the set of positive integers

uncountable set: a set that is not countable

$\aleph_0$ (aleph null): the cardinality of a countable set

c : the cardinality of the set of real numbers

Cantor diagonalization argument: a proof technique used to show that the set of real numbers is uncountable

computable function: a function for which there is a computer program in some programming language that finds its values

uncomputable function: a function for which no computer program in a programming language exists that finds its values

continuum hypothesis: the statement that no set A exists such that $\aleph_0 < |A| < c$

matrix: a rectangular array of numbers

matrix addition: see page 188

matrix multiplication: see page 189

I_n (identity matrix of order n): the $n \times n$ matrix that has entries equal to 1 on its diagonal and 0s elsewhere

A' (transpose of A): the matrix obtained from A by interchanging the rows and columns

symmetric matrix: a matrix is symmetric if it equals its transpose

zero-one matrix: a matrix with each entry equal to either 0 or 1

$A \vee B$ (the join of A and B): see page 191

$A \wedge B$ (the meet of A and B): see page 191

$A \odot B$ (the Boolean product of A and B): see page 192

RESULTS

The set identities given in Table 1 in Section 2.2

The summation formulae in Table 2 in Section 2.4

The set of rational numbers is countable.

The set of real numbers is uncountable.

TABLE 1 Set Identities.

<i>Identity</i>	<i>Name</i>
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{\overline{A}} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap C$ $A \cap (B \cup C) = (A \cap B) \cup C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \overline{A} \cup \overline{B}$ $\overline{A \cup B} = \overline{A} \cap \overline{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \overline{A} = U$ $A \cap \overline{A} = \emptyset$	Complement laws

TABLE 2 Some Useful Summation Formulae.

<i>Sum</i>	<i>Closed Form</i>
$\sum_{k=0}^n ar^k \ (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, r \neq 1$
$\sum_{k=1}^n k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^n k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^n k^3$	$\frac{n^2(n+1)^2}{4}$
$\sum_{k=0}^{\infty} x^k, x < 1$	$\frac{1}{1-x}$
$\sum_{k=1}^{\infty} kx^{k-1}, x < 1$	$\frac{1}{(1-x)^2}$

- The set of rational numbers is **countable**.
- The set of real numbers is **uncountable**.